

PathPilot: An AI-Powered Career Recommendation and Resume Intelligence System for Engineering Students

PHASE I REPORT

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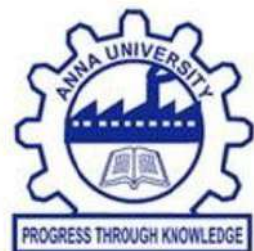
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BONAFIDE CERTIFICATE

Certified that this Report titled “**PathPilot: An AI-Powered Career Recommendation and Resume Intelligence System for Engineering Students**” is the bonafide work of **PAVITHRA N (230701225), NAVITHA M (230701208), NEHA K (230701209), NEHAA SESH S (230701210)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

This project addresses the following Sustainable Development Goal(s): Goal – **SDG 4 -Quality Education: Supports students in gaining relevant skills and knowledge through personalized career guidance for better education and future employment.**

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ABSTRACT

Choosing the right career path and preparing for campus placements can be challenging for college students, particularly when they lack proper guidance about the skills, qualifications, and preparation required for their desired roles. Students often depend on multiple platforms for career information, resume preparation, skill development, and interview practice, making the overall preparation process time-consuming and difficult to manage. To address these challenges, **PathPilot – An AI-Powered Career Guidance and Placement Preparation Platform** is developed to provide students with a centralized and personalized career preparation environment.

The proposed system integrates five major modules: AI Career Recommendation, Trending Skills, Resume Analyzer, Interview Preparation, and AI Career Assistant. Based on the student's profile, educational background, skills, and career interests, the platform helps identify suitable career opportunities and relevant skills to pursue. The Resume Analyzer assists students in reviewing their resumes against career requirements, while the Interview Preparation module provides aptitude assessments, technical tests, coding resources, HR and behavioral interview practice, communication preparation, and AI-based mock interviews. The Trending Skills module highlights skills associated with different career roles, helping students understand what they should learn to move towards their desired profession.

The AI Career Assistant, powered by the Gemini API, uses a limited, domain-specific knowledge base focused primarily on Computer Science and Engineering (CSE) career and placement information. It retrieves relevant knowledge for a user's query and generates a response using the available context. When sufficient information is unavailable, the system is designed to indicate this limitation rather than provide an unsupported answer. The platform is implemented using HTML, CSS, and JavaScript for the frontend, Python Flask for backend development, and SQLite for database management. Flask session-based authentication is used to protect access to user-specific features, while a daily progress tracking facility encourages students to maintain consistent preparation habits.

Currently, PathPilot focuses on CSE students, with scope for future expansion to other engineering and IT departments, a broader knowledge base, and enhanced career guidance capabilities. By bringing essential career preparation activities together in one application, PathPilot aims to help students move from career exploration to placement readiness through a more organized and accessible process.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
AI	Artificial Intelligence
API	Application Programming Interface
ATS	Applicant Tracking System
CSE	Computer Science and Engineering
DBMS	Database Management System
DOCX	Microsoft Word Document Format
Flask	Python web framework used for the application server
Gemini	Generative AI model/service used for selected AI tasks
HR	Human Resources
JSON	JavaScript Object Notation
MCQ	Multiple Choice Question
NLP	Natural Language Processing
OS	Operating System
PDF	Portable Document Format
SQL	Structured Query Language
UI	User Interface
UX	User Experience

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Choosing an appropriate career is one of the most important decisions in a student's life. With the rapid growth of technology and the increasing number of professional opportunities, students often find it difficult to determine which career best matches their skills, qualifications, interests, and goals.

Traditional career guidance generally depends on personal counselling, internet searches, recommendations from friends or family, and general information available through educational websites. These approaches may not always provide personalized recommendations based on the individual's profile.

The Continue Career project is developed as a web-based career guidance system that uses a structured career dataset to assist users in exploring suitable career opportunities. The application is developed using Flask, which provides the backend functionality required for handling user inputs and processing career-related data.

The system allows users to provide relevant information and receive career recommendations based on the available dataset. It provides a simple and accessible approach to career exploration.

1.2 OBJECTIVE

The main objective of PathPilot is to provide a centralized platform that helps college students explore career opportunities and prepare for campus placements in a more organized manner.

The specific objectives are:

1. **To recommend suitable careers:** Help students identify potential career roles based on their educational background, skills, experience, projects, certifications, and interests.
2. **To identify trending skills:** Provide information about skills associated with different career roles so that students can understand what they need to learn or improve.
3. **To analyze resumes:** Help students review their resumes against relevant job-role requirements and identify areas that may need improvement.
4. **To support interview preparation:** Provide resources and practice activities for aptitude, technical subjects, coding resources, HR interviews, behavioral questions, communication, and mock interviews.
5. **To provide AI-based career assistance:** Integrate the Gemini API with a domain-specific knowledge base to answer relevant career and placement-related questions. When sufficient information is unavailable, the assistant should indicate this limitation.
6. **To encourage consistent preparation:** Include daily progress tracking so that students can monitor their preparation activities and build regular study habits.

7. **To simplify career preparation:** Bring essential career guidance and placement preparation features together in a single application, reducing the need to depend on multiple separate platforms.

1.3 EXISTING SYSTEM

In the existing approach, college students depend on various websites, online learning platforms, job portals, and career guidance applications to prepare for their future careers and campus placements. Although these platforms provide useful information, students often need to access multiple services to complete different stages of their preparation. Career exploration, skill development, resume preparation, aptitude practice, and interview guidance are generally handled through separate tools, making the overall process difficult to organize.

Traditional career guidance methods mainly depend on information provided by placement officers, faculty members, career counsellors, and online resources. While these sources offer valuable support, students may not always receive continuous, individualized guidance based on their skills, academic background, projects, certifications, and career interests. As a result, some students find it challenging to identify suitable job roles and understand the requirements for entering their preferred career fields.

Existing career recommendation platforms help users explore job opportunities and professional pathways. However, the recommendations and information available may vary depending on the platform and the details provided by the user. Students may still need to research the skills, qualifications, and technologies required for a particular role independently. Similarly, resume-building tools and resume analysis applications focus mainly on reviewing resume content or matching it with job descriptions, while interview preparation platforms provide separate resources for aptitude tests, technical questions, coding practice, and HR interviews.

Another limitation is the difficulty of obtaining career-related answers from a single source. General-purpose AI assistants can provide information about job roles, skills, and interviews, but their responses may not always be specific to a student's placement requirements. The information may also require further verification to ensure that it is relevant and reliable. Furthermore, students may need separate applications or manual methods to monitor their preparation activities and maintain consistency.

Therefore, the existing approach lacks a single, integrated environment that brings the major stages of career preparation together. Students must spend additional time collecting information, identifying learning priorities, and managing their preparation across different platforms. These limitations highlight the need for a centralized career guidance and placement preparation system that considers student profiles, provides relevant career information, supports resume improvement and interview practice, and encourages regular preparation. PathPilot is proposed to address these challenges by combining these essential features within one web-based application.

1.4 PROPOSED SYSTEM

The proposed system, **PathPilot** – An AI-Powered Career Guidance and Placement Preparation Platform, is designed to help college students make informed career decisions and prepare for campus placements through a single, user-friendly application. Instead of depending on multiple platforms for different preparation activities, students can access essential career guidance and placement resources in one place.

The system begins with user registration and login. New users can create a profile by entering details such as their educational background, department, year of study, skills, experience, certifications, projects, and career interests. These details help the platform provide career-related suggestions that are relevant to the student's profile and goals. Session-based authentication is implemented using Flask to protect user-specific features.

PathPilot consists of five major modules: AI Career Recommendation, Trending Skills, Resume Analyzer, Interview Preparation, and AI Career Assistant. The AI Career Recommendation module helps students explore suitable job roles based on their profiles and interests. The Trending Skills module highlights skills associated with different career roles, helping students identify areas for further learning. The Resume Analyzer reviews a student's resume against relevant career requirements and points out areas for improvement.

The Interview Preparation module provides resources and practice activities for aptitude, technical subjects, coding resources, HR interviews, behavioral questions, communication preparation, and AI-based mock interviews. These features help students practice different aspects of the placement process. The AI Career Assistant, named PathPilot AI, is integrated with the Gemini API and a domain-specific knowledge base focused mainly on Computer Science and Engineering careers and placement preparation. It retrieves relevant information to answer user queries and indicates when sufficient information is unavailable.

The platform also includes a daily progress tracking feature that allows students to monitor their preparation activities and encourages consistent practice. The application is developed using HTML, CSS, and JavaScript for the frontend, Python Flask for the backend, and SQLite for database management.

The current implementation primarily supports CSE students, with the possibility of expanding the knowledge base to cover other departments in the future. Overall, the proposed system aims to make career exploration and placement preparation more accessible, personalized, and organized by integrating important preparation activities into one platform.

CHAPTER 2 – LITERATURE SURVEY

2.1 LITERATURE REVIEW

The development of artificial intelligence and web-based applications has created new opportunities for improving career guidance and placement preparation for college students. Several studies have explored career recommendation systems, resume analysis, job matching, and AI-based counselling. These studies provide useful insights into how technology can help students understand career opportunities and prepare for employment.

Lahoud et al. (2023) presented a comparative analysis of recommender systems for university major selection and career-domain guidance. Their study examined different approaches, including collaborative filtering, demographic-based recommendations, knowledge-based reasoning, and ontology-based methods. The research highlighted the importance of considering students' characteristics and available career information when suggesting suitable academic and professional pathways. This work provides a foundation for developing systems that help students explore career options based on their profiles.

The study “Trends and Characteristics of Career Recommendation Systems for Fresh Graduated Students” (2022) reviewed research on career recommendation systems designed for recent graduates. It examined how user profiles, job information, and text-mining techniques are used to recommend appropriate career opportunities. The review identified hybrid recommendation approaches as a commonly used method in the studies examined. Its findings emphasize the importance of combining relevant student information with job-related requirements when designing career recommendation systems.

Modak et al. (2024) reviewed approaches for resume analysis and job-description matching using machine learning and natural language processing. Their work discussed techniques for extracting information from resumes, identifying relevant skills, and comparing candidate profiles with job requirements. The review demonstrated the importance of understanding resume content and matching it with employment expectations. These concepts are relevant to developing resume analysis features that help students identify missing information and improve their job-readiness.

Sarmurzin et al. (2026) conducted a systematic review of artificial intelligence in career counselling for university students. The study examined research on AI-powered chatbots, predictive analytics, and recommendation systems used in higher education. It discussed how conversational agents can make career information more accessible while also highlighting concerns related to trust, transparency, and the reliability of AI-generated guidance. The review supports the use of AI as an assistance tool while emphasizing the importance of responsible and relevant responses.

From the reviewed studies, it can be understood that technology can support different stages of career preparation, including career selection, skill identification, resume improvement, and career-related question answering. However, these areas are often addressed through separate tools or specialized systems. Students may therefore need to switch between multiple platforms to complete their placement preparation.

PathPilot is proposed to address this practical gap by integrating five modules into a single application: AI Career Recommendation, Trending Skills, Resume Analyzer, Interview Preparation, and AI Career Assistant. Unlike systems that depend on a trained machine-learning recommendation

model, the current PathPilot implementation uses student-profile information and predefined career knowledge for career suggestions, while the AI assistant uses relevant domain-specific knowledge with the Gemini API. The platform primarily focuses on CSE students and aims to make career exploration and placement preparation more organized and accessible.

2.2 COMPARISON AND DISCUSSION

Existing career guidance and placement preparation systems provide students with different resources, such as job recommendations, resume-building tools, online courses, aptitude tests, and interview preparation materials. However, these services are often available on separate platforms, requiring students to search for information and manage their preparation independently.

PathPilot brings these important activities together in a single application. The system provides five major modules: AI Career Recommendation, Trending Skills, Resume Analyzer, Interview Preparation, and AI Career Assistant. These modules are designed to help students explore suitable career opportunities, identify relevant skills, improve their resumes, practise for interviews, and obtain career-related guidance.

The following table presents a comparison between the general existing approach and the proposed PathPilot system.

S. No.	Comparison Factor	Existing System	Proposed PathPilot System
1	Career Guidance	Students often depend on separate career guidance platforms.	Provides career recommendations based on student profile information and career interests.
2	Skill Identification	Students search different websites to identify job-related skills.	Highlights trending skills associated with different career roles.
3	Resume Analysis	Students may use separate resume-building or checking tools.	Reviews resumes against relevant career requirements and identifies areas for improvement.
4	Interview Preparation	Preparation resources may be distributed across multiple platforms.	Integrates aptitude, technical, HR, behavioral, communication, and mock interview preparation features.
5	Career Assistance	Students may search online or use general-purpose AI assistants.	Provides a domain-specific AI career assistant using relevant knowledge and the Gemini API.
6	Personalization	Guidance may require students to interpret information and select suitable resources themselves.	Uses profile details and career interests to provide relevant career-related suggestions.

7	Progress Tracking	Students may track their preparation manually or through separate tools.	Includes a daily progress tracking feature to monitor preparation activities.
8	Accessibility of Resources	Students may need to switch between different websites and applications.	Combines major career guidance and placement preparation features in one platform.

The comparison shows that PathPilot aims to reduce the effort involved in managing different stages of career preparation. Its integrated approach allows students to explore career options, understand the skills required for particular roles, review their resumes, and practise for placement interviews within the same application.

The AI Career Assistant further supports students by retrieving relevant information from a domain-specific knowledge base and using the Gemini API to generate responses. When sufficient information is unavailable in the knowledge base, the assistant is designed to indicate this limitation rather than provide unsupported information.

Although PathPilot combines several useful features, its current implementation primarily focuses on Computer Science and Engineering (CSE) students. The quality of recommendations and AI-generated responses depends on the available career knowledge, profile information, and external AI service. Therefore, broader department coverage, systematic evaluation, and continuous knowledge-base updates can be considered for future improvement.

Overall, the proposed system provides a centralized and organized approach to career guidance and placement preparation. It aims to help students make informed career decisions and prepare more effectively by bringing essential resources together in one application.

2.3 CONCLUSION

The literature survey highlights the importance of technology in supporting students with career selection and placement preparation. The reviewed studies discuss different approaches to career recommendation, resume analysis, job matching, and AI-based career guidance. These approaches show how digital tools can help students understand job requirements and prepare for employment opportunities.

However, students may still need to depend on multiple platforms to access career information, identify relevant skills, analyse their resumes, and practise for interviews. Managing these activities separately can make the preparation process difficult and time-consuming.

To address this challenge, PathPilot integrates career recommendation, trending skills, resume analysis, interview preparation, and AI-based career assistance into a single platform. It uses student profile information and career-related knowledge to provide relevant guidance, while the Gemini API supports responses through the AI Career Assistant.

The current system focuses primarily on CSE students and provides a foundation for further development. In the future, the platform can be expanded to support additional academic departments, improve the coverage of its knowledge base, and undergo more extensive evaluation.

Thus, PathPilot aims to make career guidance and placement preparation more accessible, structured, and convenient for college students.

CHAPTER 3 – SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The system architecture of PathPilot illustrates how the different components of the application work together to support students throughout their career preparation journey. The system follows a structured workflow that begins with user registration or login and continues through the dashboard, career guidance modules, and daily progress tracking.

The process starts at the landing page, where users can explore the overview, features, and career-related information available on the platform. New users can register and complete their profiles by providing details such as their educational background, skills, projects, certifications, and career interests. Existing users can log in directly. Flask session-based authentication is used to manage user sessions and restrict access to the application's protected features.

After successful login, users are directed to the home dashboard, which provides access to the five major modules of PathPilot.

The **AI Career Recommendation** module uses the available profile details and career interests to suggest suitable job roles and provide related information, including required skills and other role-specific details. The **AI Career Assistant**, named PathPilot AI, allows students to ask career and placement-related questions. It retrieves relevant information from the available domain-specific knowledge base and uses the Gemini API to generate responses. If sufficient information is unavailable, the assistant indicates this limitation.

The **Resume Analyzer** module allows students to upload their resumes for analysis. It extracts relevant information, such as skills, education, projects, and experience, and compares the resume with relevant career requirements to provide feedback and improvement suggestions. The **Interview Preparation** module supports preparation through aptitude and technical questions, subject- or role-based practice, and AI-based mock interviews. It also provides feedback to help students identify areas for improvement.

The **Trending Skills** module presents in-demand skills and related technologies associated with different career roles. It helps students understand which skills may be useful for their career interests and identify suitable learning resources.

The activities performed across these modules contribute to the application's daily progress tracking feature, which helps students monitor their preparation activities, including learning and practice. This encourages students to maintain consistent preparation habits and work towards their career goals.

The frontend is developed using HTML, CSS, and JavaScript, while Python Flask manages the backend operations. SQLite is used for database management, and the Gemini API supports AI-generated responses in the career assistant. Overall, the architecture integrates the major features of PathPilot

into a single application, making career exploration and placement preparation more organized and accessible to students.

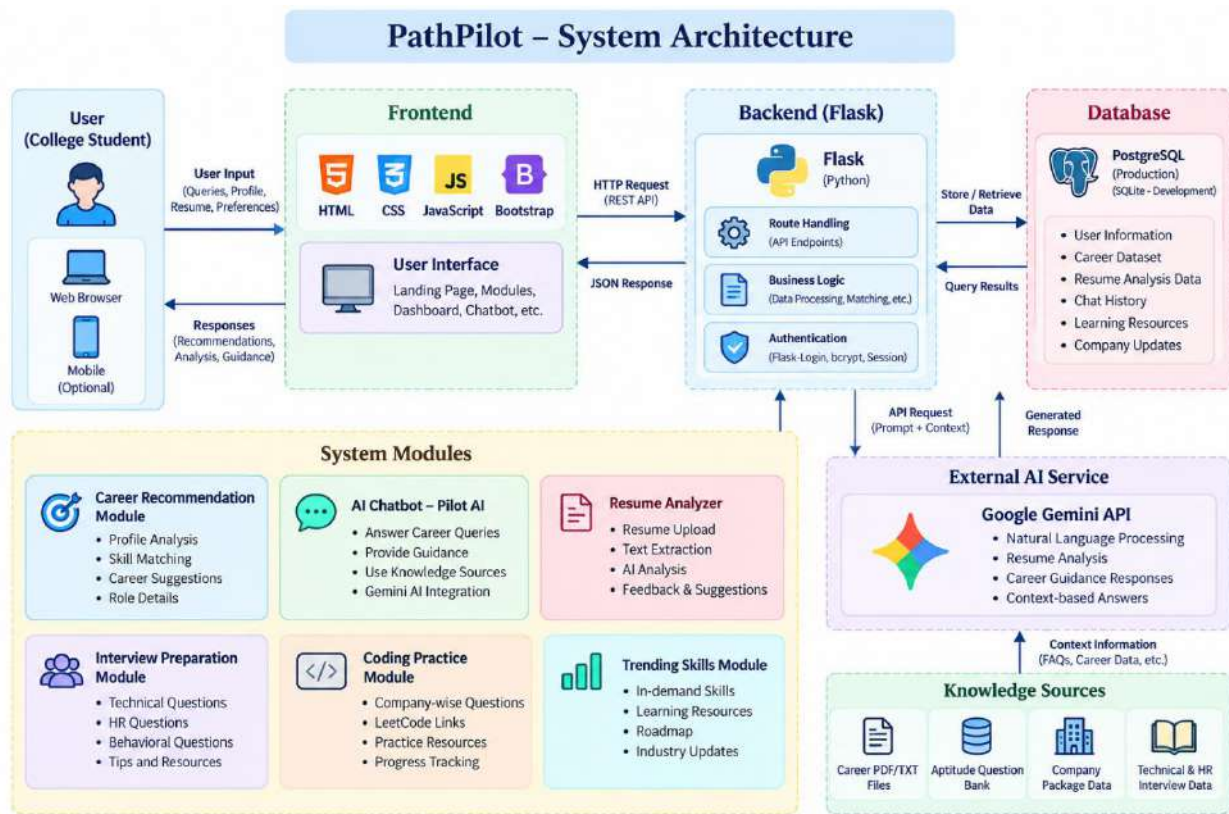


Figure 4.1: System Architecture of the Continue Career System

3.2 SYSTEM REQUIREMENTS

System requirements describe the software and hardware resources needed to develop and run the PathPilot application. The platform combines a web-based user interface, a Flask backend, an SQLite database, and the Gemini API for AI-based career assistance. A computer with a modern web browser and a stable internet connection is sufficient for accessing the application and its online AI functionality.

The system requirements are divided into software requirements and hardware requirements. These requirements help ensure that the application runs properly and that users can access its career guidance and placement preparation features.

WORKFLOW

The system workflow begins when users enter their profile details, skills, interests, or upload their resumes through the web interface. The Flask backend processes the information, interacts with the database, AI services, and knowledge sources, and generates personalized career recommendations, resume analysis, chatbot responses, and interview preparation resources.

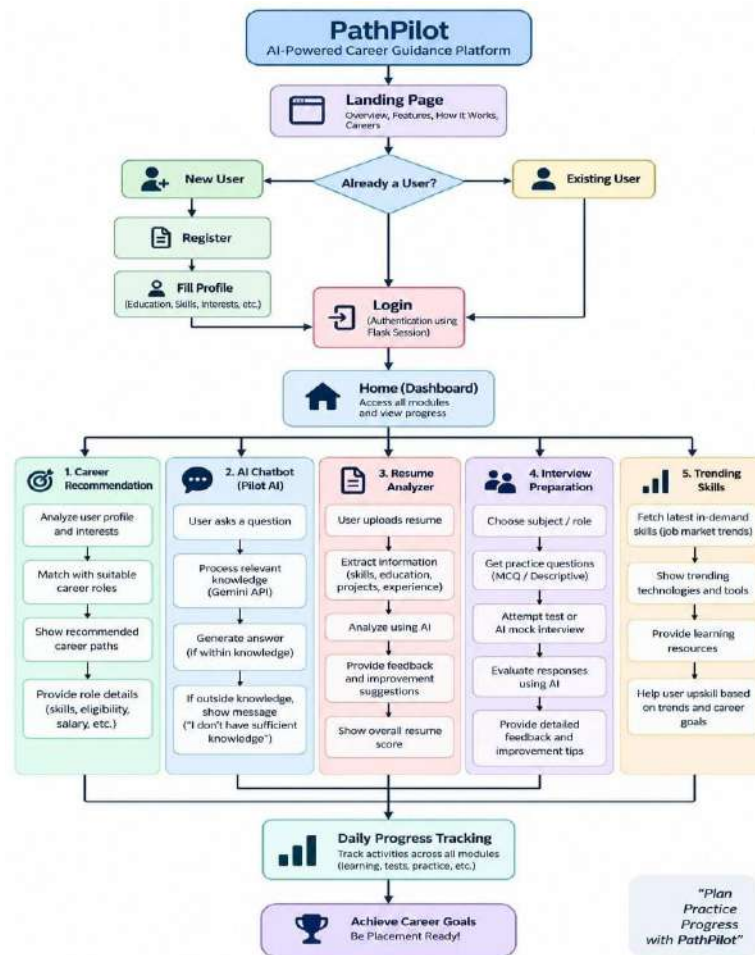


Figure 3.1: System Architecture and Workflow of PathPilot

3.2.1 SOFTWARE REQUIREMENTS

Operating System

Windows 10/11, Linux, or macOS can be used to develop and run the application.

Programming Language

Python is used for backend development, application logic, and data processing.

Backend Framework

Flask is used as the web application framework for handling routing, requests, responses, session management, and backend operations.

Frontend Technologies

HTML

CSS

JavaScript

These technologies are used to design the user interface, apply styling, and implement interactive features.

Database

SQLite is used to store user profiles, application data, and other information required by the system.

AI Service

Gemini API is integrated into the PathPilot AI Career Assistant to generate responses to career-related queries using relevant domain-specific knowledge.

Development Environment

Visual Studio Code is used to write, edit, and manage the application code.

Python Package Manager

pip is used to install and manage the Python packages required for the application.

Web Browser

Google Chrome, Microsoft Edge, or another modern web browser is used to access and test the application.

Supporting Libraries

Python libraries are used according to the implemented features, including libraries for API integration, data processing, and resume analysis.

Internet Connection

An internet connection is required to communicate with the Gemini API and access online resources used by the application.

3.2.2 HARDWARE REQUIREMENTS

The proposed system does not require specialized hardware.

Minimum Requirements

Processor: Intel Core i3 or equivalent

RAM: 4 GB

Storage: Minimum 10 GB available space

Display: 1366 × 768 resolution or above

Network: Stable internet connection

Recommended Requirements

Processor: Intel Core i5 or equivalent

RAM: 8 GB or above

Storage: 256 GB SSD or above

The system can run effectively on standard computers and laptops.

3.2.3 DATASET REQUIREMENTS

The career dataset is one of the major components of the Continue Career system. It contains structured information related to different career opportunities.

The dataset may contain attributes such as:

Career/Job Title

Required Skills

Educational Qualification

Technical Skills

Job Description

Experience Requirements

Career Category

Related Career Roles

The dataset should be properly structured and cleaned before being used by the application. Missing values, duplicate records, inconsistent formats, and irrelevant fields should be handled during preprocessing.

The dataset is used by the application to identify career options that are relevant to the user's input.

3.2.4 DEPLOYMENT AND SCALING

The PathPilot application can initially be deployed in a local development environment for testing and validation. After successful testing, it can be hosted on a suitable web server or cloud platform to make it accessible to users.

Deployment Process

The deployment process follows these stages:

Development → Testing → Production Deployment → Monitoring

Development: Develop the application using Python Flask, the frontend technologies, and SQLite.

Testing: Verify the application's modules, authentication, database operations, and AI integration.

Production Deployment: Host the application in a suitable Python-compatible hosting environment.

Monitoring: Monitor application performance, errors, API availability, and user activity.

The application can be deployed using a hosting environment that supports Flask and the required dependencies.

Scalability

Scalability refers to the ability of the application to support future growth in the number of users, data volume, and application workload. PathPilot can be extended through the following improvements:

1. Database Optimization: Use an appropriate production database when the volume of stored data and concurrent users increases.
2. Data Processing Optimization: Retrieve and process only the relevant career knowledge required for each query.
3. Caching Mechanisms: Cache frequently accessed information to reduce repeated processing.
4. Cloud Deployment: Host the application on a cloud platform to support increased accessibility and resource requirements.
5. Frontend and Backend Separation: Separate application components when required to simplify maintenance and independent scaling.
6. AI Service Optimization: Improve knowledge retrieval and manage API requests efficiently as usage increases.
7. Load Balancing: Distribute incoming requests across multiple application instances when traffic becomes high.

These improvements provide a foundation for expanding PathPilot in the future. The current implementation can be enhanced gradually based on user demand and performance requirements.

3.3 SYSTEM DESIGN

The system design of PathPilot describes how the different components of the application interact to provide career guidance and placement preparation for college students. It explains how user information is collected, processed, stored, and used to provide relevant results through the application's modules.

The system begins when a user opens the PathPilot website. New users can register and provide their profile details, while existing users can log in using their credentials. Flask manages the authentication process through session management and allows authorized users to access the dashboard and its features.

The frontend is developed using HTML, CSS, and JavaScript to provide a user-friendly interface. When a student selects a feature or submits information, the request is sent to the Python Flask backend for processing. The backend manages application logic, validates user input, communicates with the database, and coordinates the operations required by each module.

SQLite is used to store user profiles and other application data. The AI Career Recommendation module uses available profile details and career-related knowledge to suggest suitable roles. The Trending Skills module presents relevant skills and technologies associated with career opportunities. The Resume Analyzer processes uploaded resumes and provides feedback based on relevant career requirements, while the Interview Preparation module provides practice activities and interview-related resources.

The AI Career Assistant, known as PathPilot AI, uses the Gemini API along with relevant information retrieved from a domain-specific knowledge base. This helps generate responses to career and placement-related queries. If sufficient information is unavailable, the assistant indicates the limitation rather than providing an unsupported answer.

The Daily Progress Tracking feature records relevant preparation activities and helps students monitor their engagement with the platform. Overall, the system design connects the frontend, backend, database, AI service, and career preparation modules to provide an organized environment for students to explore career opportunities and prepare for placements.

3.3.1 ACTIVITY DIAGRAM

The activity diagram illustrates the sequence of activities performed by users while interacting with the PathPilot application. It describes the flow from opening the application and registering or logging in to accessing the dashboard and using the available modules. It also represents how user requests are processed by the Flask backend, how data is accessed from the database, and how the Gemini API supports AI-generated responses. Finally, the diagram shows how daily preparation activities are recorded through the progress tracking feature.

Activity Diagram – PathPilot

User Interaction Flow

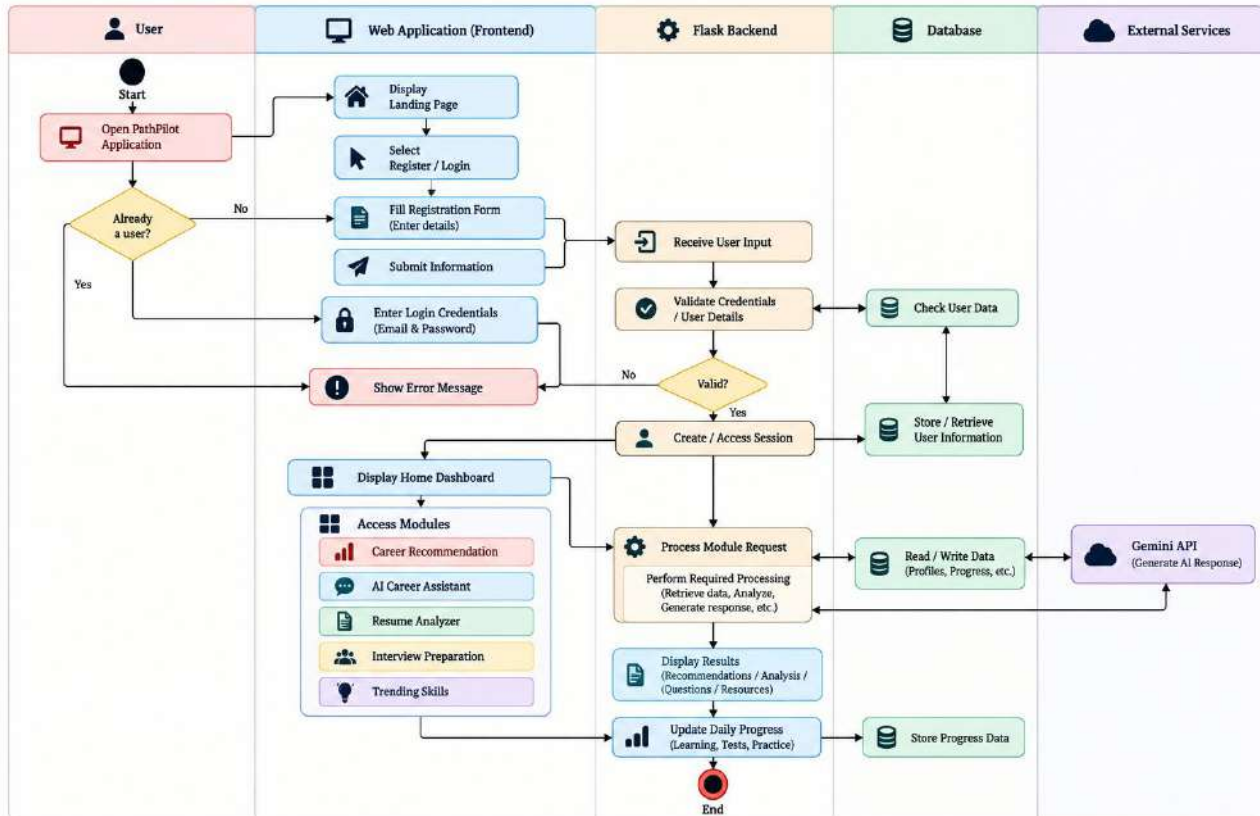


Figure 3.2: Activity Diagram of PathPilot

The figure represents the interaction between the user, web application, Flask backend, database, and external AI service. It provides a clear overview of how the different components coordinate to support career recommendations, resume analysis, interview preparation, trending skills, and AI-based career assistance.

3.3.2 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) illustrates how data moves through the Continue Career System, from user input to the generation of personalized career recommendations. It represents the interaction between the user, Flask web application, career dataset, and recommendation module.

3.3.2 DATA FLOW DIAGRAM

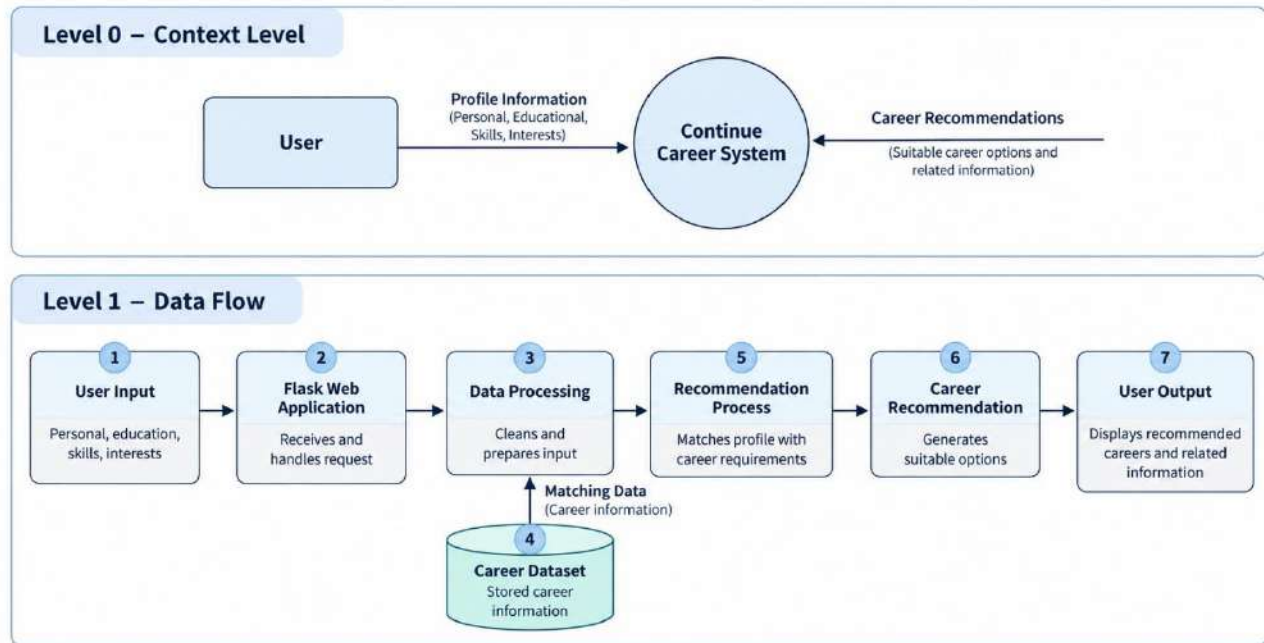


Figure 3.3.2: Data Flow Diagram of the Continue Career System

Figure description:

The diagram consists of two levels:

Level 0 – Context Level: This level provides an overview of the system's interaction with the user. The user submits personal, educational, skill, and interest information to the Continue Career System, which generates suitable career recommendations.

Level 1 – Data Flow: This level explains the internal flow of data through seven stages: user input, Flask web application, data processing, career dataset, recommendation process, career recommendation, and user output. The system processes the user's profile, matches it with stored career information, and displays relevant career options.

4. PROJECT DESCRIPTION

4.1 METHODOLOGIES

PathPilot is an AI-powered career guidance platform designed to help students identify suitable career opportunities, analyze their resumes, understand skill gaps, and receive personalized career guidance. The system combines student profile information, career-related datasets, resume information, and artificial intelligence to provide personalized recommendations.

The methodology of PathPilot begins with student registration and profile creation. The student provides educational information, skills, proficiency levels, projects, certifications, experience, and career interests. This information is stored in the database and is later used as contextual information for the AI-based modules.

The system mainly follows the process of collecting user information, storing the information, processing career-related data, analyzing uploaded resumes, and generating personalized recommendations using an AI model.

The major workflow of the system is:

Student Registration → Profile Creation → Data Storage → AI Processing → Career Guidance and Resume Analysis

PathPilot consists of two major AI-based modules. The first module is the AI Career Guidance Chatbot, which provides career-related guidance based on the student's profile and available career knowledge. The second module is the Resume Analyzer, which analyzes the uploaded resume by comparing it with the student's profile, skills, career interests, and target role.

The career knowledge base contains information related to skills, job roles, companies, packages, certifications, GCC hiring, and career mapping. This knowledge is used to improve the relevance of AI-generated responses.

The overall methodology allows PathPilot to provide personalized career guidance instead of providing the same generic recommendations to every student.

4.2 MODULES

PathPilot is divided into several interconnected modules. Each module performs a specific function while working together to provide a complete career guidance platform.

4.2.1 USER AND PROFILE MANAGEMENT MODULE

The User and Profile Management Module is responsible for managing student registration, login, authentication, and profile information.

During registration, the student creates an account by providing the required details. After successful login, the system creates a user session using the unique user ID. The session is used to identify the currently logged-in student.

The profile module stores important student information such as:

- Name
- Educational qualification
- Department
- College
- Year of study
- Skills
- Skill proficiency

- Projects
- Certifications
- Experience
- Career interests

CAREER RECOMMENDATION MODULE

The Career Recommendation Module is one of the core modules of the Continue Career System. It helps students identify suitable career opportunities based on their educational qualifications, technical skills, interests, and career preferences.

The module collects the user's profile information and processes it using the Flask backend. The system compares the user's skills and interests with the available career dataset to identify relevant career paths. Based on this matching process, suitable career recommendations are displayed along with related information to help students make informed career decisions.

Key functionalities:

- **Profile Analysis:** Collects and processes the user's educational details, skills, and interests.
- **Career Dataset Integration:** Accesses stored career information, including relevant skills and career requirements.
- **Skill Matching:** Compares the user's profile with the requirements of different career paths.
- **Personalized Recommendations:** Displays suitable career options based on the matching results.
- **Career Guidance:** Helps students understand potential career paths and identify skills relevant to their preferred roles.

The module provides a centralized approach to career exploration, enabling students to make better-informed decisions about their professional development.

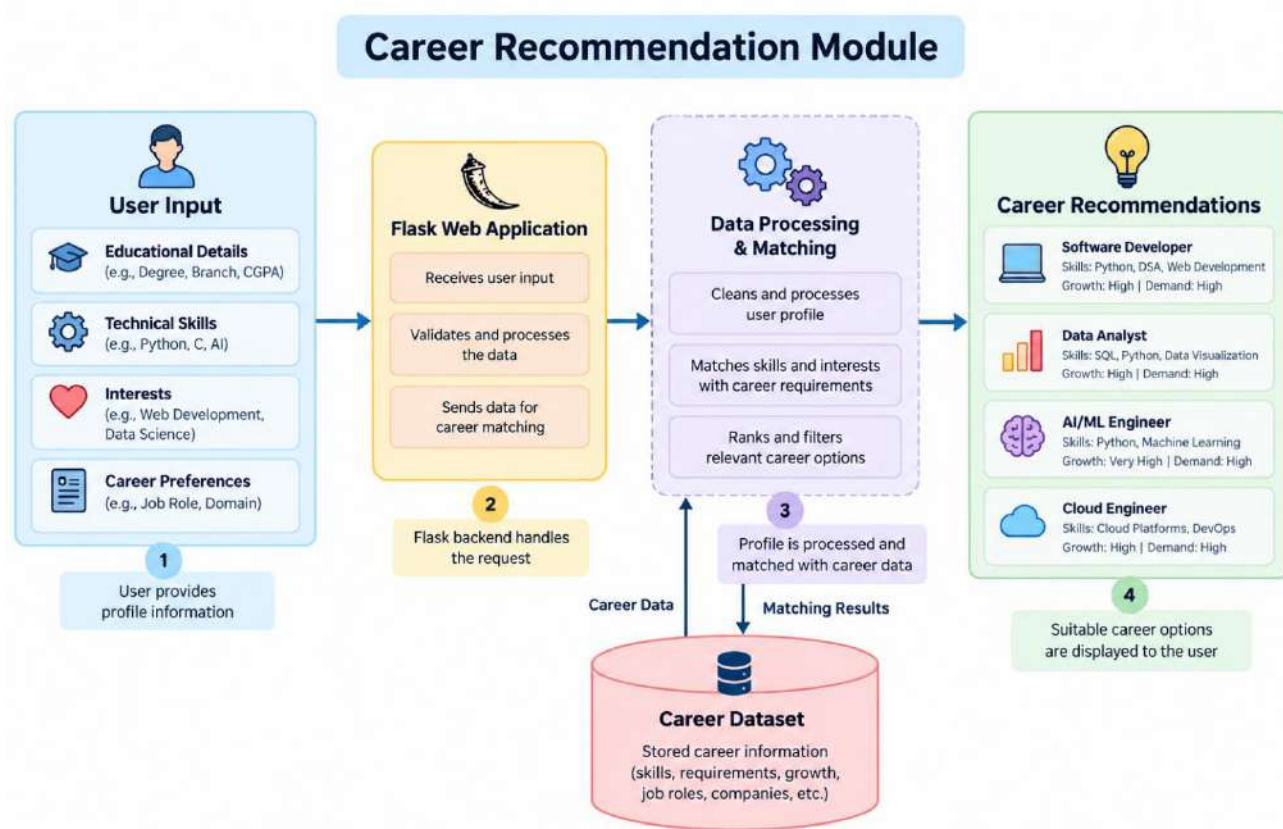


Figure 4.2.1: Career Recommendation Module

4.2.2 AI CAREER GUIDANCE CHATBOT MODULE

The AI Chatbot – Pilot AI Module provides an interactive interface through which users can ask questions related to career opportunities, technical skills, interview preparation, resume improvement, and learning resources. It uses the Flask backend to process user queries and integrates with the Gemini AI model to generate relevant responses using available knowledge sources.

The chatbot can provide guidance related to:

- Career paths
- Job roles
- Required skills
- Companies
- Certifications
- Learning recommendations
- Career planning
- Skill development
- Career opportunities

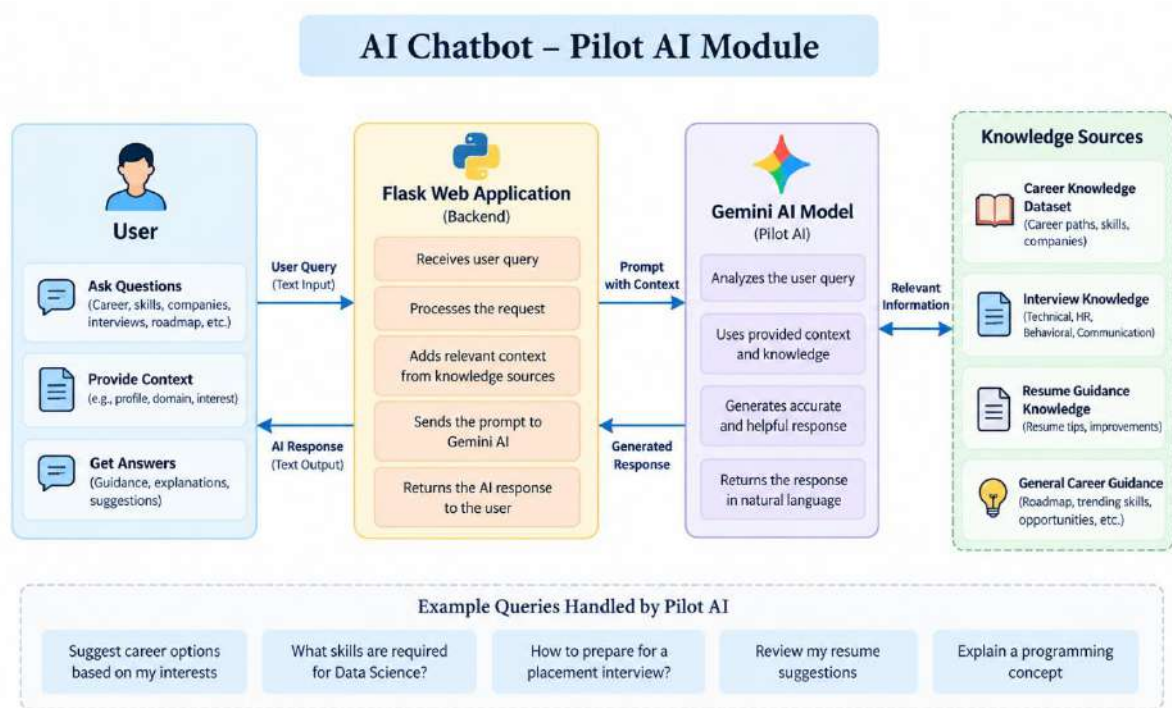


Figure 4.2.2: AI Chatbot – Pilot AI Module

The career knowledge base is loaded by the Flask application during startup. The system can use information from PDF, TXT, and Excel datasets. The Excel dataset contains information related to company packages, global certifications, GCC COE hiring, and career mapping.

The student profile is used to make the chatbot response more relevant to the individual student.

For example, a student with Python, SQL, and Machine Learning skills can receive different career guidance from a student who has Java, Spring Boot, and SQL skills.

Therefore, the chatbot acts as a personalized AI career assistant rather than a general-purpose chatbot.

4.2.3 RESUME ANALYZER MODULE

The Resume Analyzer is the second major AI-based module of PathPilot. It is designed to analyze a student's resume and provide personalized feedback. The student uploads a resume through the Resume Analyzer interface. The system extracts the text from the uploaded document and retrieves the student's profile information from the database.

The system can consider the following information:

- Resume content
- Student profile
- Skills
- Skill proficiency

- Education
- Projects
- Certifications
- Experience
- Career interests
- Target job role
- Job description

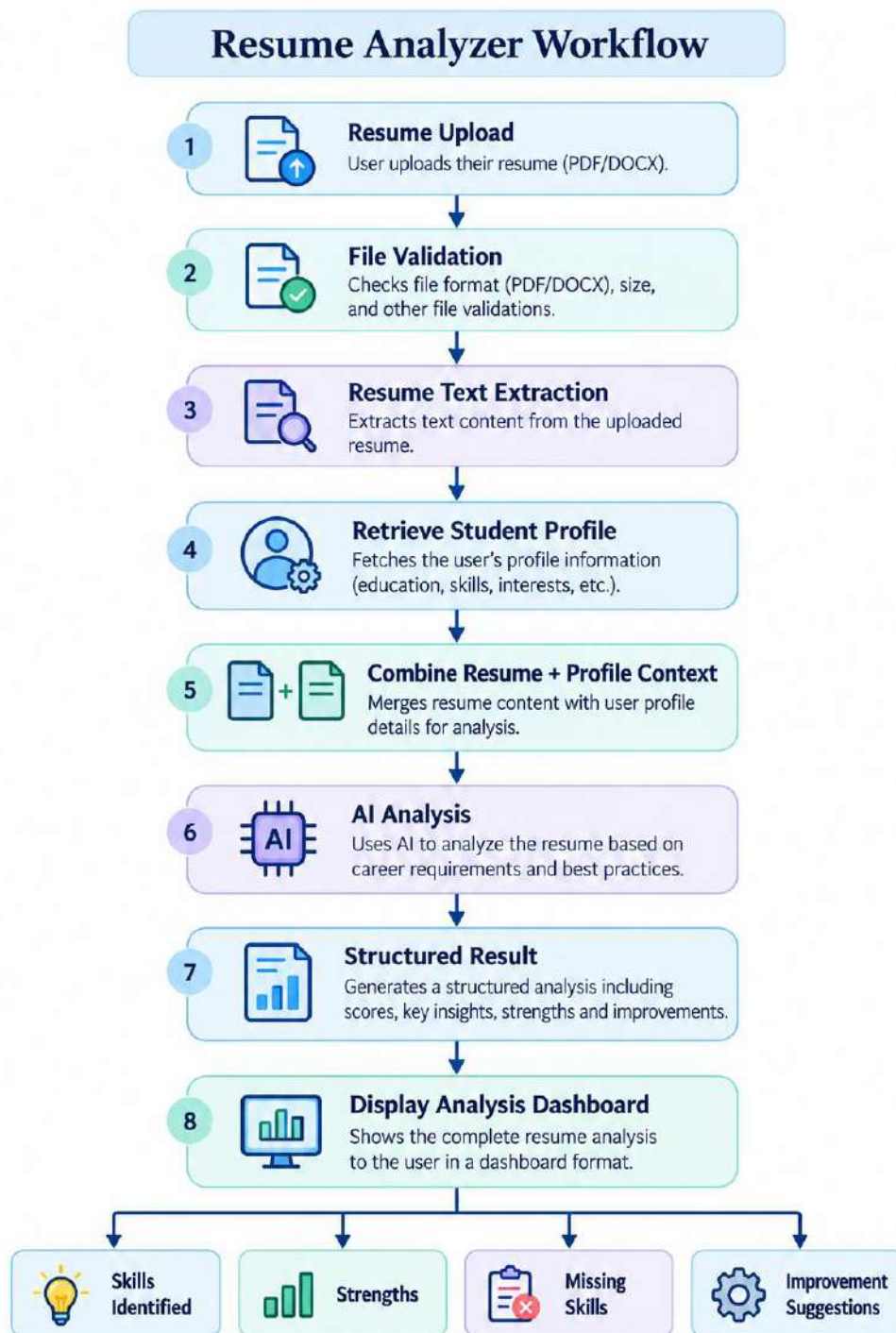


Figure 4.2.3: Resume Analyzer Workflow

The AI analysis can generate information such as:

- Overall resume score
- ATS-related score
- Resume strengths
- Resume weaknesses
- Matched skills
- Missing keywords
- Skill gaps
- Improvement suggestions
- Recommended skills
- Suitable job roles
- Suitable companies
- Next steps

One of the important concepts in the Resume Analyzer is the distinction between a missing resume keyword and an actual skill gap.

For example, if Python is already present in the student's profile but is not mentioned in the uploaded resume, the system can identify Python as a missing resume keyword. However, if the student does not have a particular skill in their profile or resume, that skill can be considered as a potential skill gap.

Therefore, the Resume Analyzer provides more personalized feedback by using the student's existing profile information together with the resume.

4.2.4 CAREER KNOWLEDGE BASE MODULE

The Career Knowledge Base Module provides career-related information to the AI modules. PathPilot uses multiple knowledge sources to improve the quality and relevance of the generated responses.

The knowledge base includes:

- Skill-to-job-role information
- Company career guidance information
- Resume analysis requirements
- Company package information
- Global certifications
- GCC COE hiring information
- Career mapping information

The system loads the available knowledge files from the knowledge directory when the Flask application starts.

The Excel dataset contains multiple sheets such as:

- Company Packages
- Global Certifications
- GCC COE Hiring
- Career Mapping

The information from these datasets is converted into text-based context and made available to the AI processing system. This allows the chatbot and other AI-based modules to use project-specific career information rather than relying only on general AI knowledge.

4.2.5 DATABASE AND USER INTERFACE MODULE

The Database and User Interface Module manages persistent student information and provides the web interface through which students interact with PathPilot.

SQLite is used as the database system.

The database stores information related to:

- Users
- Student profiles
- User skills
- Resume assessments

The Users table stores authentication-related information.

The Profiles table stores detailed student information. The User Skills table stores individual skills and their proficiency levels. The Assessments table stores the results generated from resume analysis. The frontend interface is developed using HTML, CSS, JavaScript, and Jinja2 templates. Flask is used to connect the frontend with the backend.

The major pages of the system include:

- Landing Page
- Registration
- Login
- Profile
- AI Career Guidance Chatbot
- Resume Analyzer

The user interface provides students with a simple way to access the different features of the PathPilot platform.

5. IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

PathPilot was implemented as a Flask-based web application that integrates student profile management, a career knowledge base, AI-powered career guidance, and resume analysis.

The backend manages user authentication, database operations, profile retrieval, resume processing, knowledge loading, and communication with the AI model.

The major implementation results are discussed below.

5.1.1 IMPLEMENTATION OF STUDENT PROFILE MANAGEMENT

The Student Profile Management module was implemented to collect and store important information about each student. The student enters information related to education, skills, skill proficiency, projects, certifications, experience, and career interests. The information is stored in the SQLite database and associated with the corresponding user ID.

When the student logs into PathPilot, the session stores the user ID. Protected routes use this user ID to retrieve the correct student's information. The profile information is then made available to the AI-based modules.

The implementation provides the following flow:

Login



Session Created



User ID Retrieved

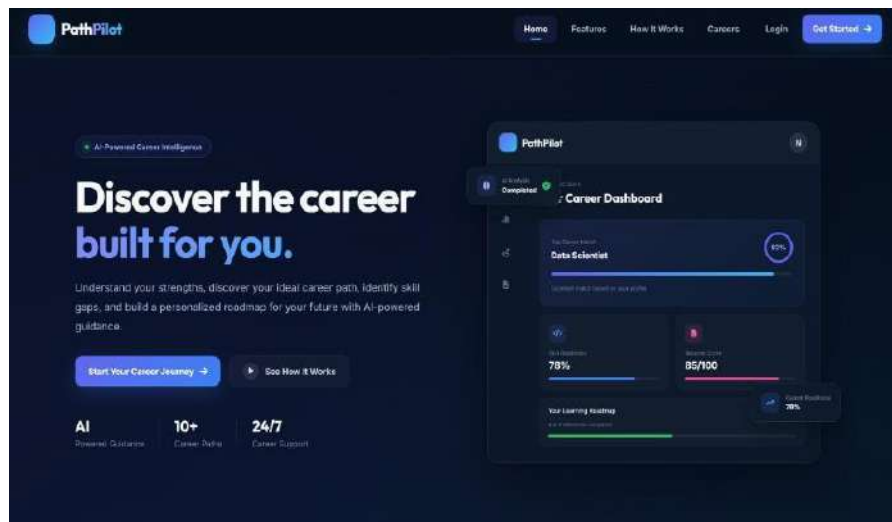


Profile Retrieved from Database



Profile Used by AI Modules

This connection allows PathPilot to provide personalized career guidance.



5.1.2 IMPLEMENTATION OF AI CAREER GUIDANCE CHATBOT

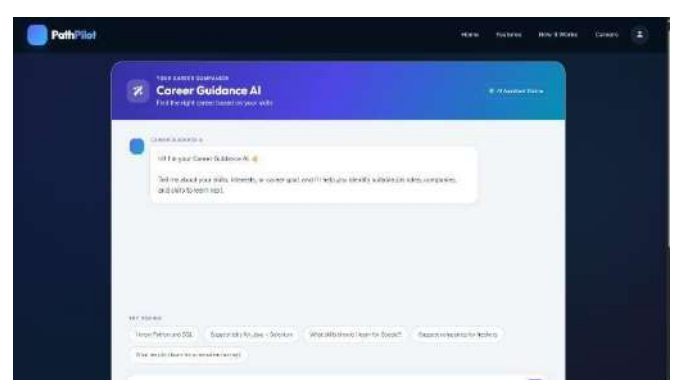
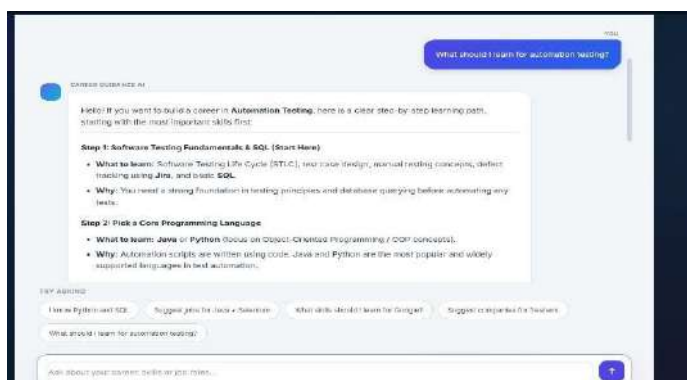
The AI Career Guidance Chatbot was implemented using the Flask backend and an AI model.

When the student enters a question, the backend receives the query and prepares the required context.

The context can include:

- Student profile information
- Student skills
- Career knowledge
- Company information
- Certification information
- User question

The backend constructs a suitable prompt and sends the information to the Gemini AI model. The generated response is then returned to the frontend and displayed in the chatbot interface.



The implementation allows the chatbot to answer career-related questions in an interactive manner.

5.1.3 IMPLEMENTATION OF RESUME ANALYSIS

The Resume Analyzer was implemented to process resumes uploaded by students and generate structured feedback.

The uploaded resume is first validated and its text is extracted.

The system then retrieves the profile of the logged-in student from the database.

The extracted resume text and profile information are combined with the target career role and job description when available.

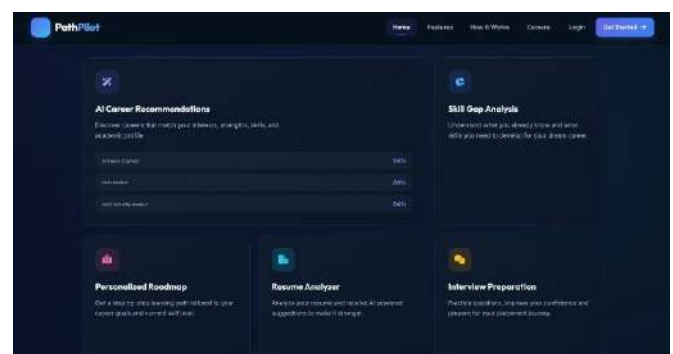
This information is passed to the AI model with instructions to analyze the resume.

The AI generates structured results that can be displayed through different components of the Resume Analyzer dashboard.

The result can contain:

- Overall Score
- ATS Score
- Strengths
- Weaknesses
- Keywords
- Skill Analysis
- Skill Gaps
- Priority Improvements
- Learning Recommendations
- Job Roles
- Companies
- Next Step

The structured result makes it possible to present the analysis using visual components such as score cards, skill tags, progress bars, comparison sections, and recommendation cards.



This approach makes the resume analysis easier for students to understand.

5.2 DISCUSSION

The implementation of PathPilot demonstrates how artificial intelligence can be integrated with student-specific information to provide personalized career guidance.

The main advantage of the system is the connection between the student profile and the AI modules.

The AI Career Guidance Chatbot can use the student's educational background, skills, interests, and other profile information while generating career-related responses.

Similarly, the Resume Analyzer does not analyze the uploaded resume in isolation. It retrieves the student's profile and compares the resume information with the available profile information.

This allows the system to identify whether a skill is actually missing or simply not mentioned in the resume.

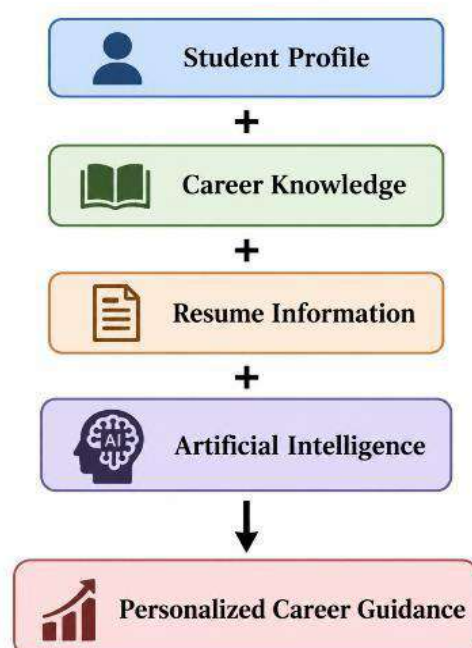
For example, if a student has Python as an existing skill in their profile but the uploaded resume does not contain Python, the system can recommend adding Python to the resume.

On the other hand, if the student does not have a required skill in the profile or resume, the system can identify it as a potential skill gap and recommend learning it.

The use of a career knowledge base further improves the usefulness of the system by providing information related to job roles, companies, packages, certifications, GCC hiring, and career mapping.

The structured AI output also makes it possible to display the results in a visual format rather than presenting a large amount of plain text.

Overall, the implementation demonstrates the integration of:



6.1 RESULT

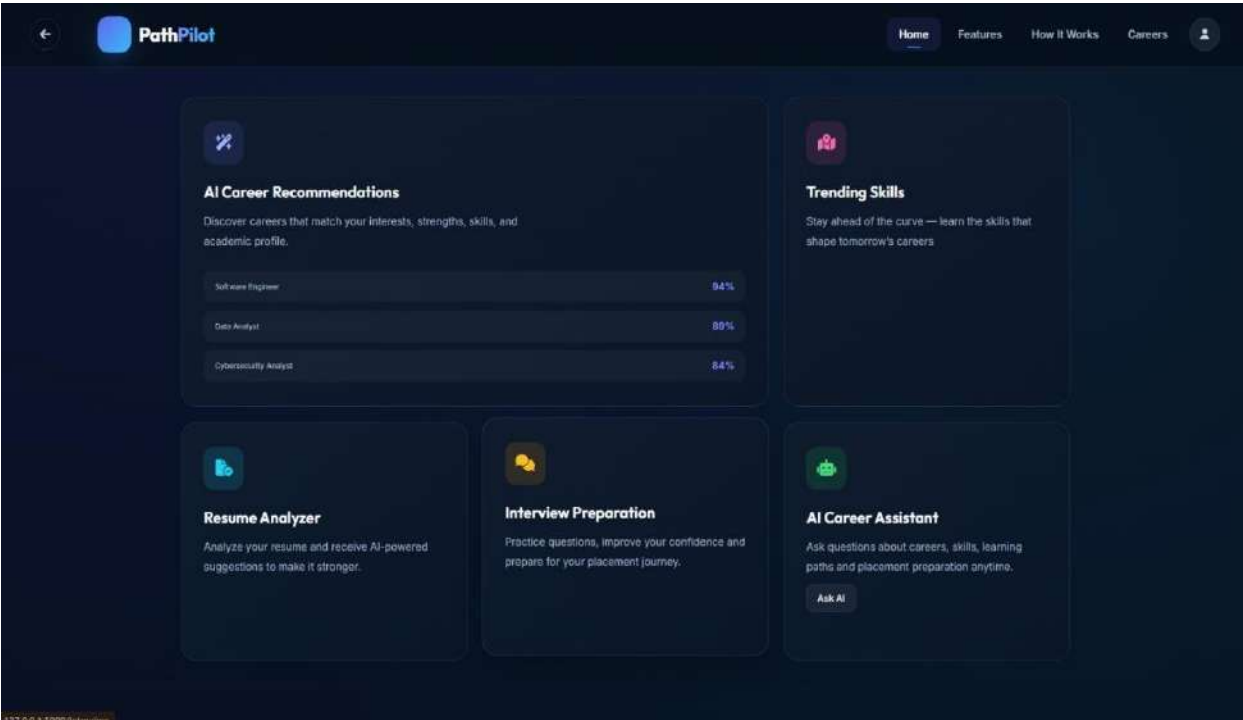


FIGURE 1 DASHBOARD

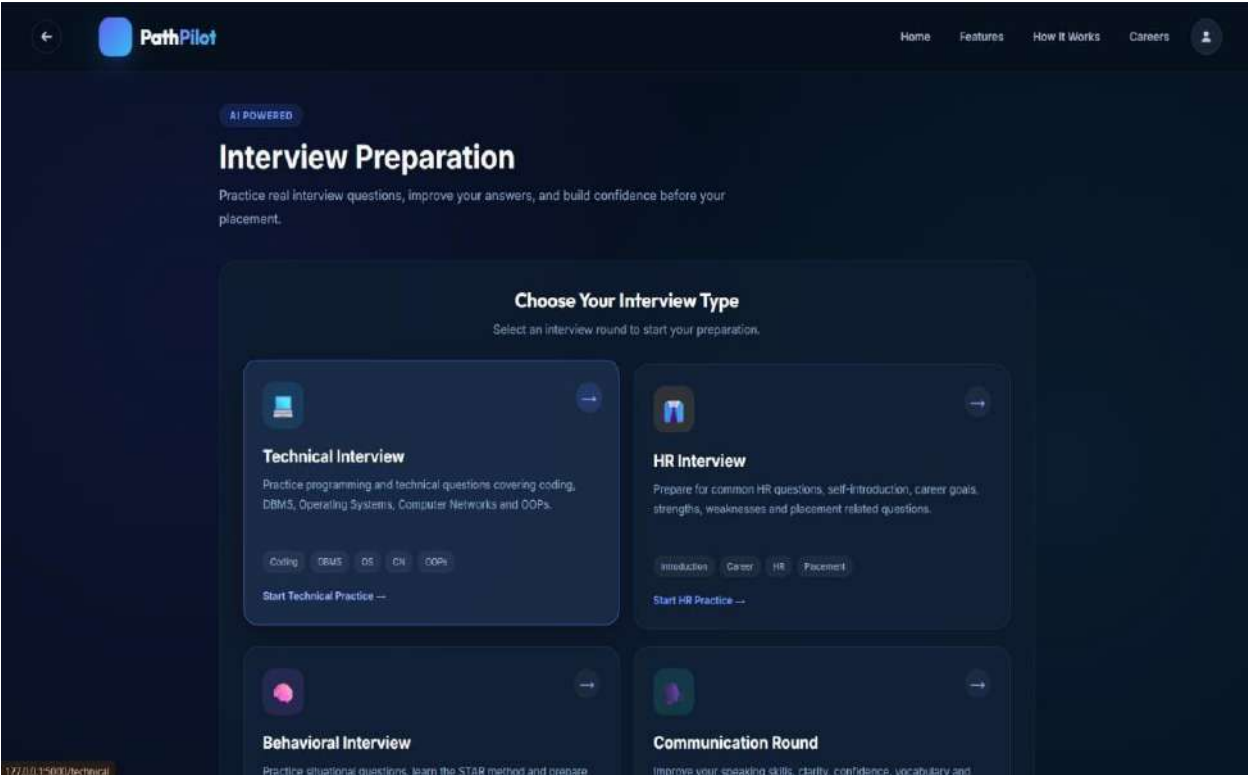


FIGURE 2. Interview Preparation – Interview Type Selection

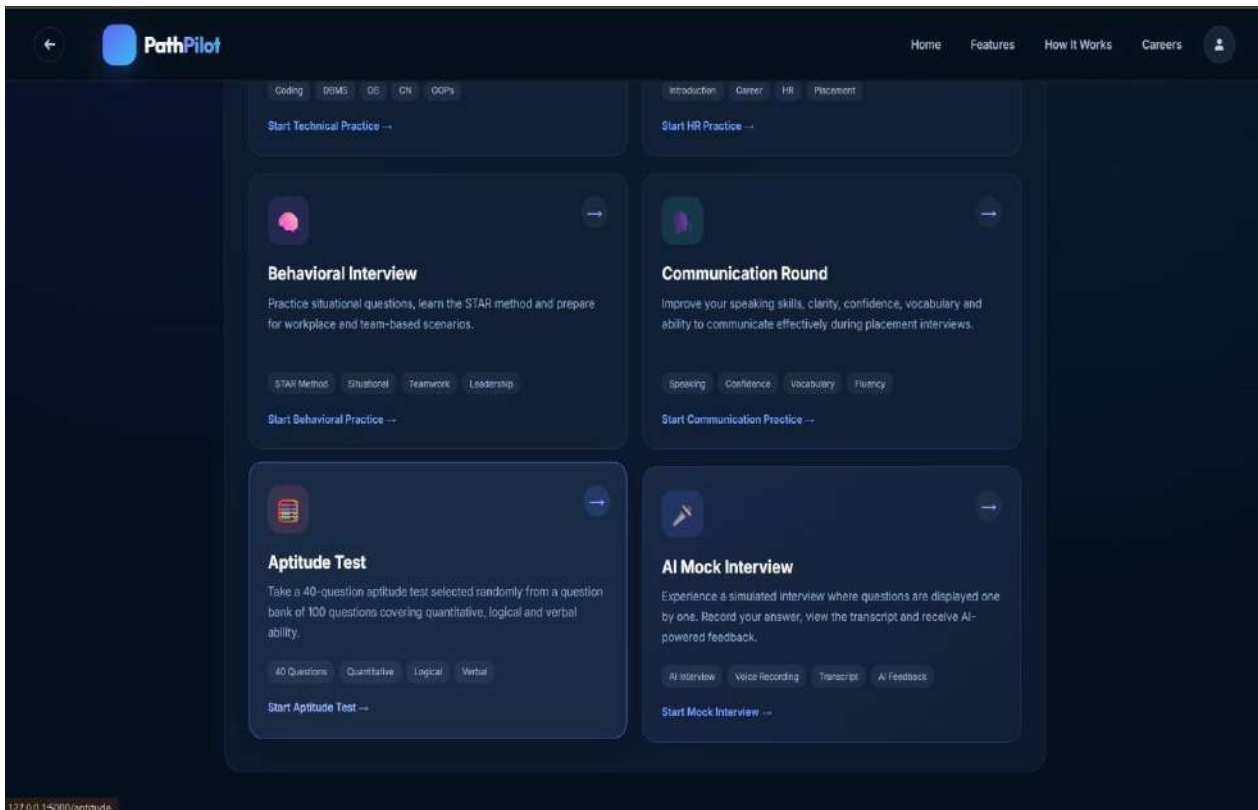


FIGURE 3. Interview Preparation – Interview Categories

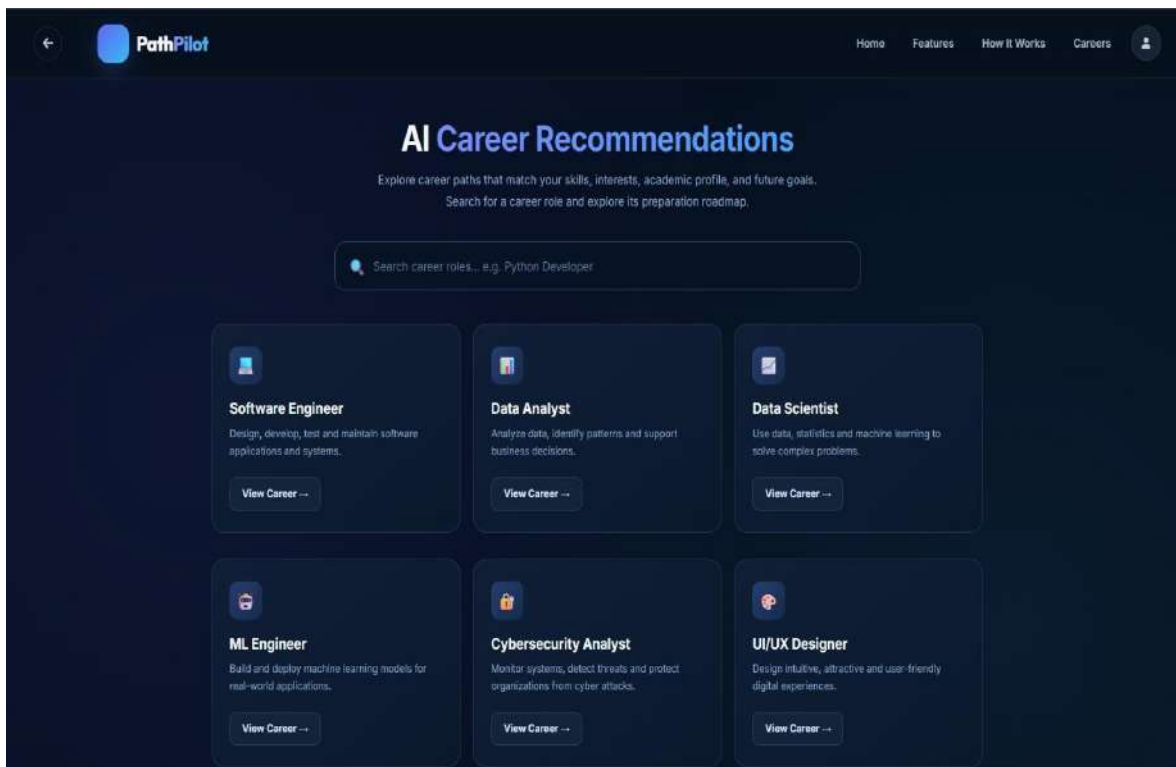


FIGURE 4. AI Career Recommendations – Career Path Selection

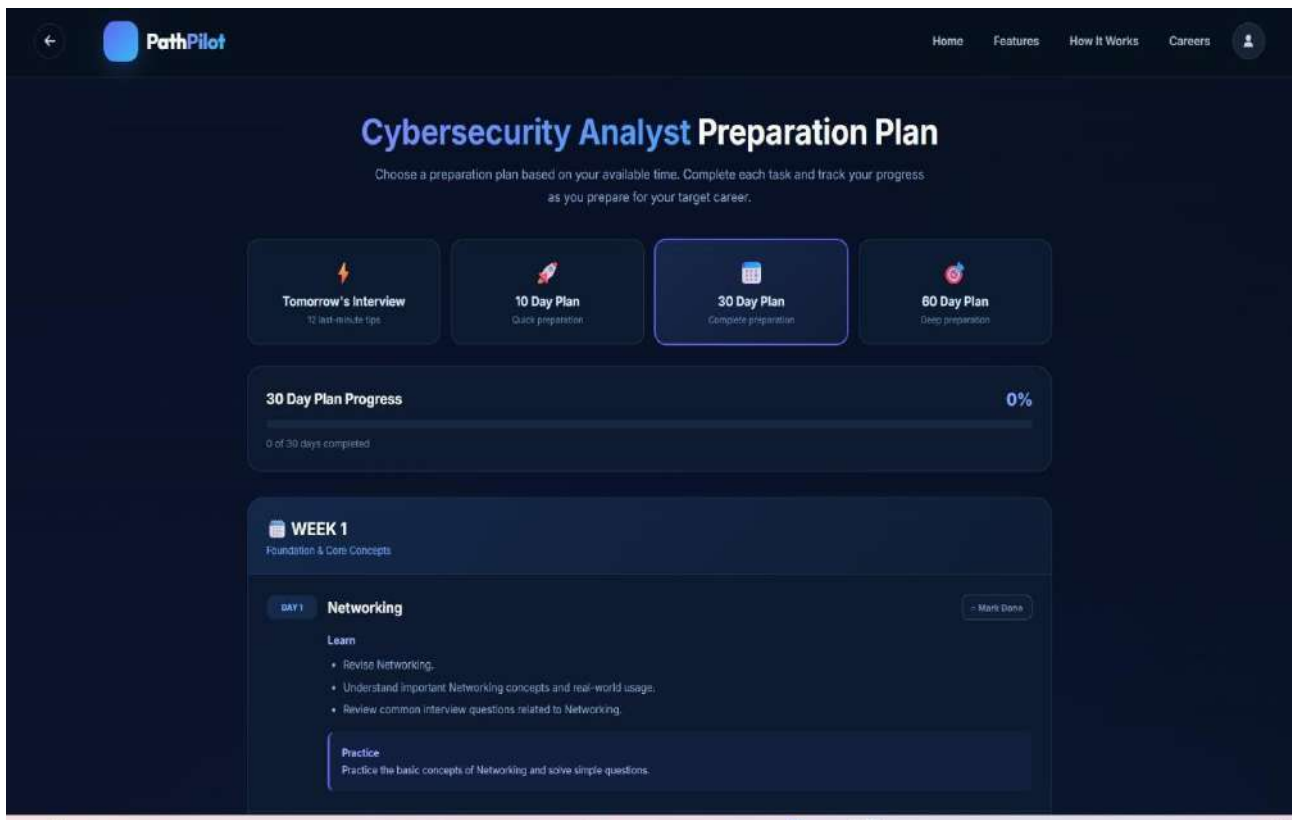


FIGURE 5. Personalized Career Preparation Plan

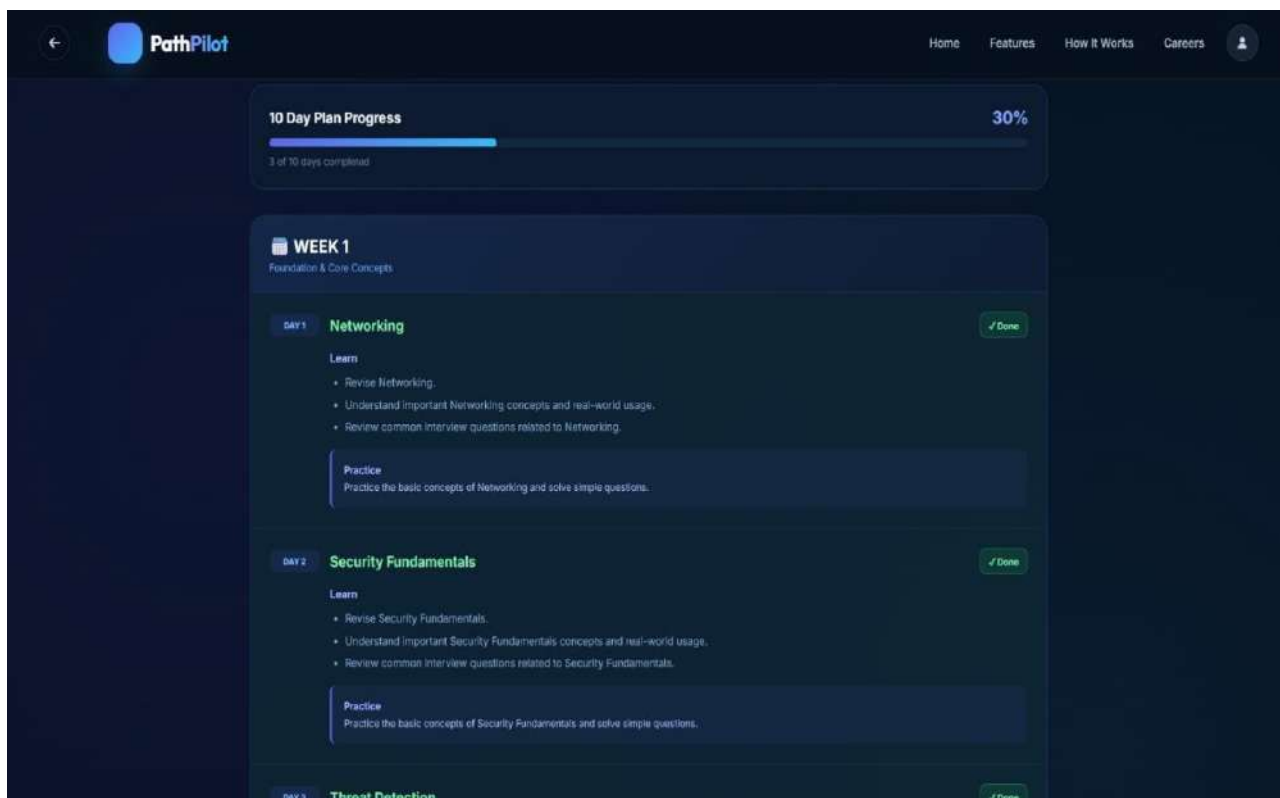


FIGURE 6. Weekly Career Preparation Progress

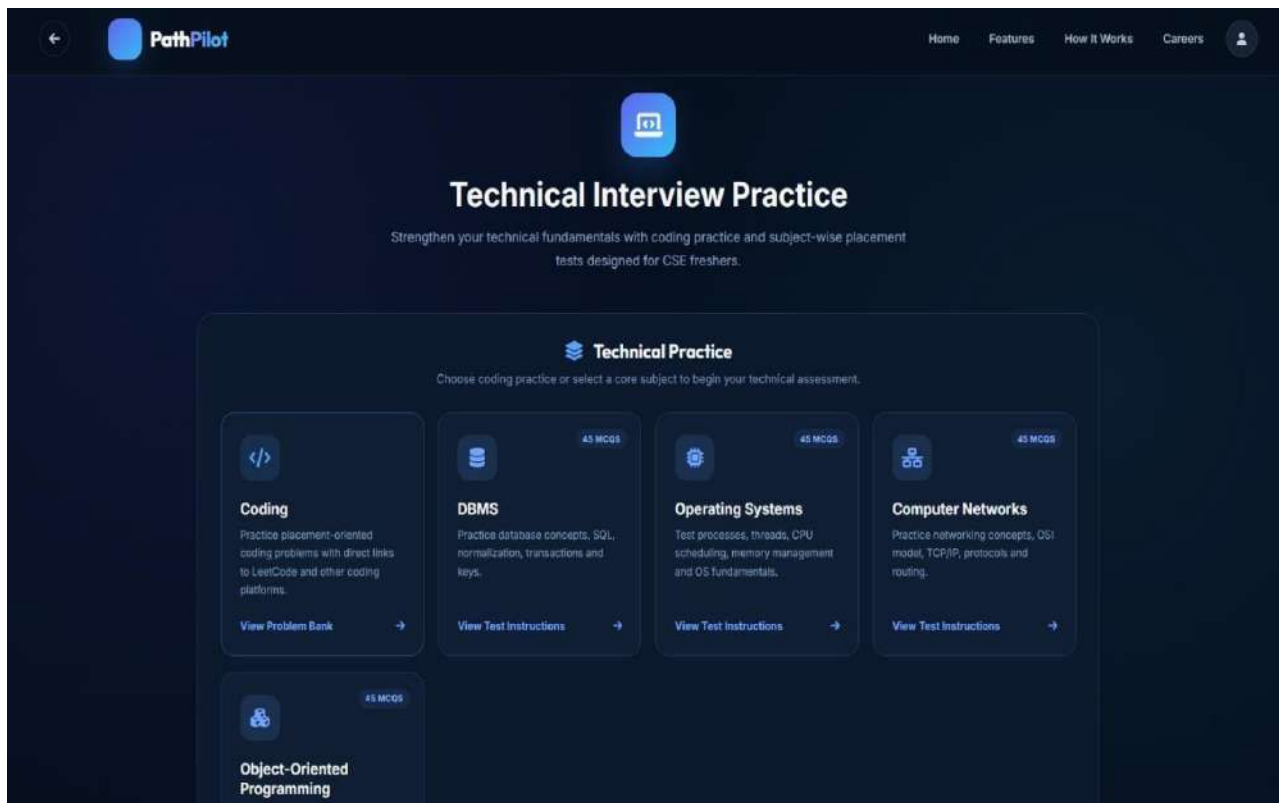


FIGURE 7. Technical Interview Practice – Topic Selection

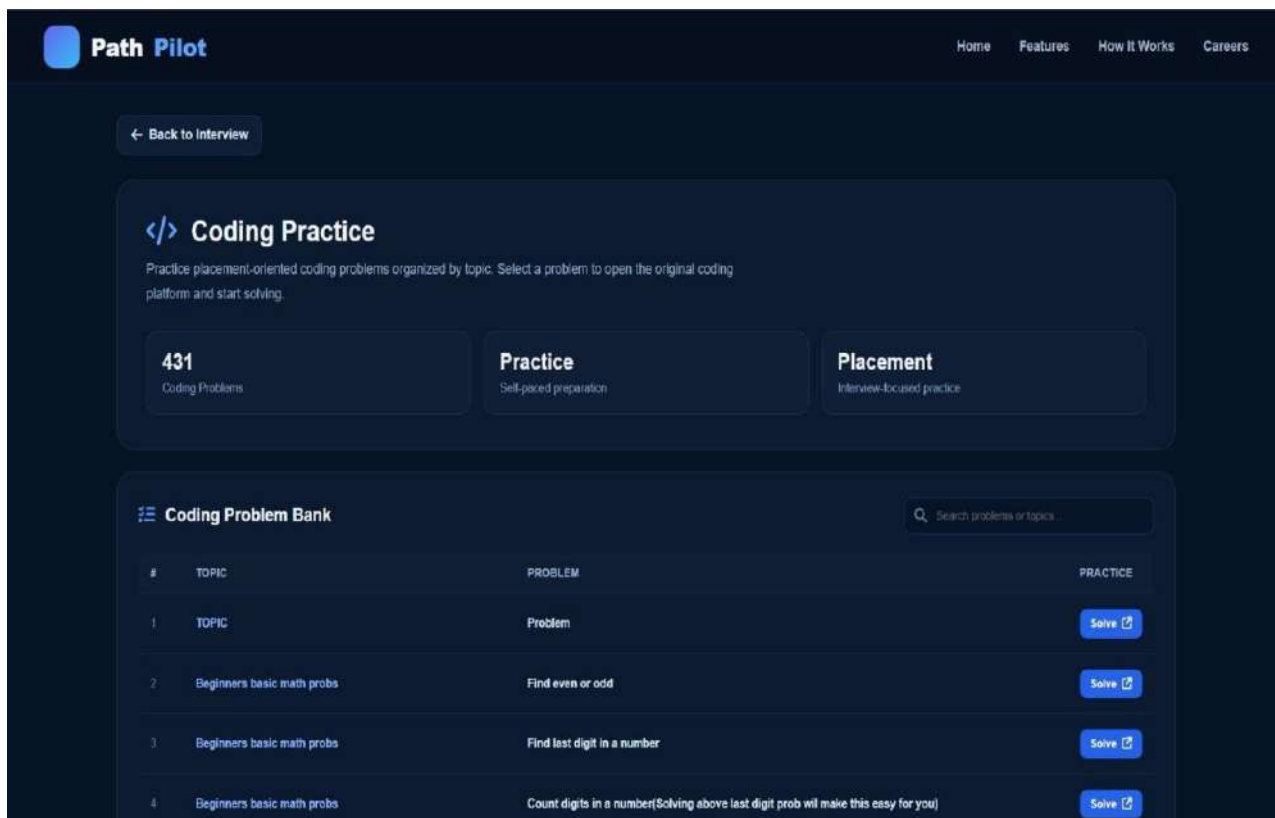


FIGURE 8. Coding Practice – Problem Bank

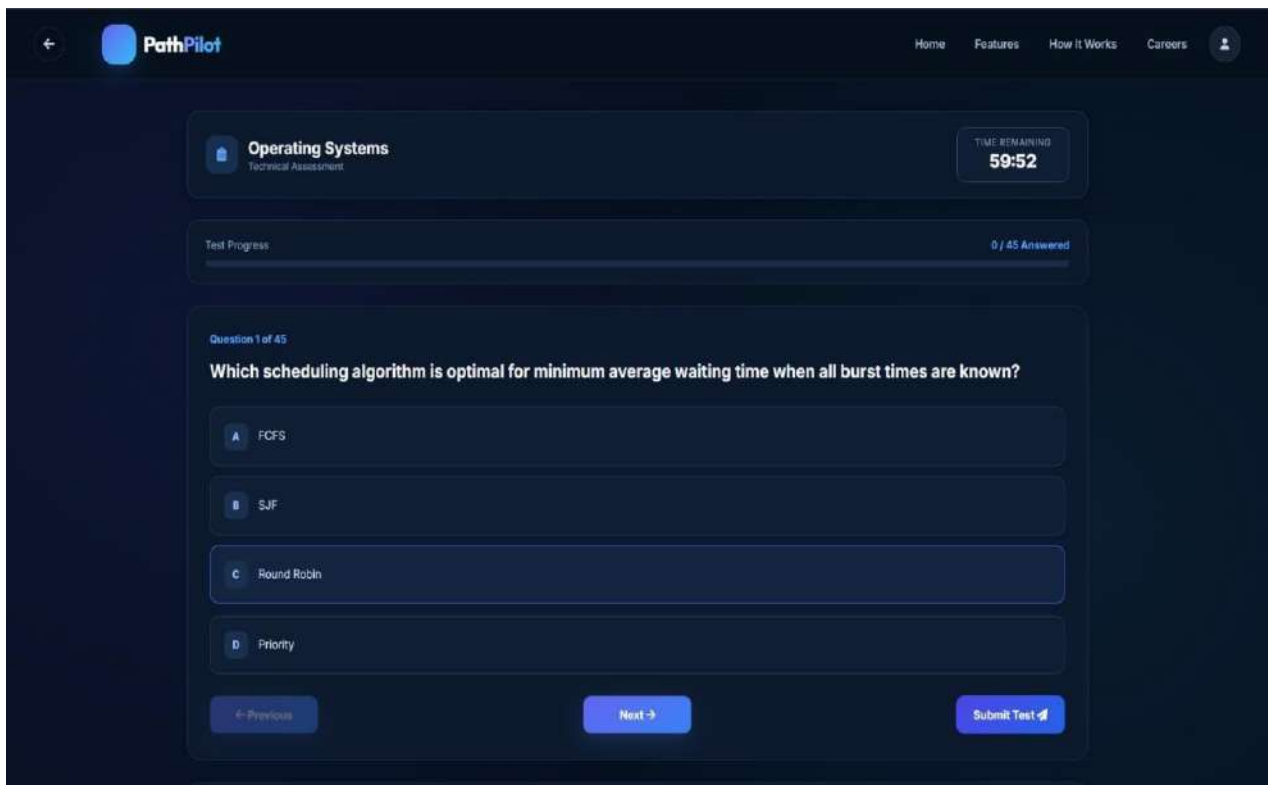


FIGURE 9. Coding Practice – MCQ Assessment

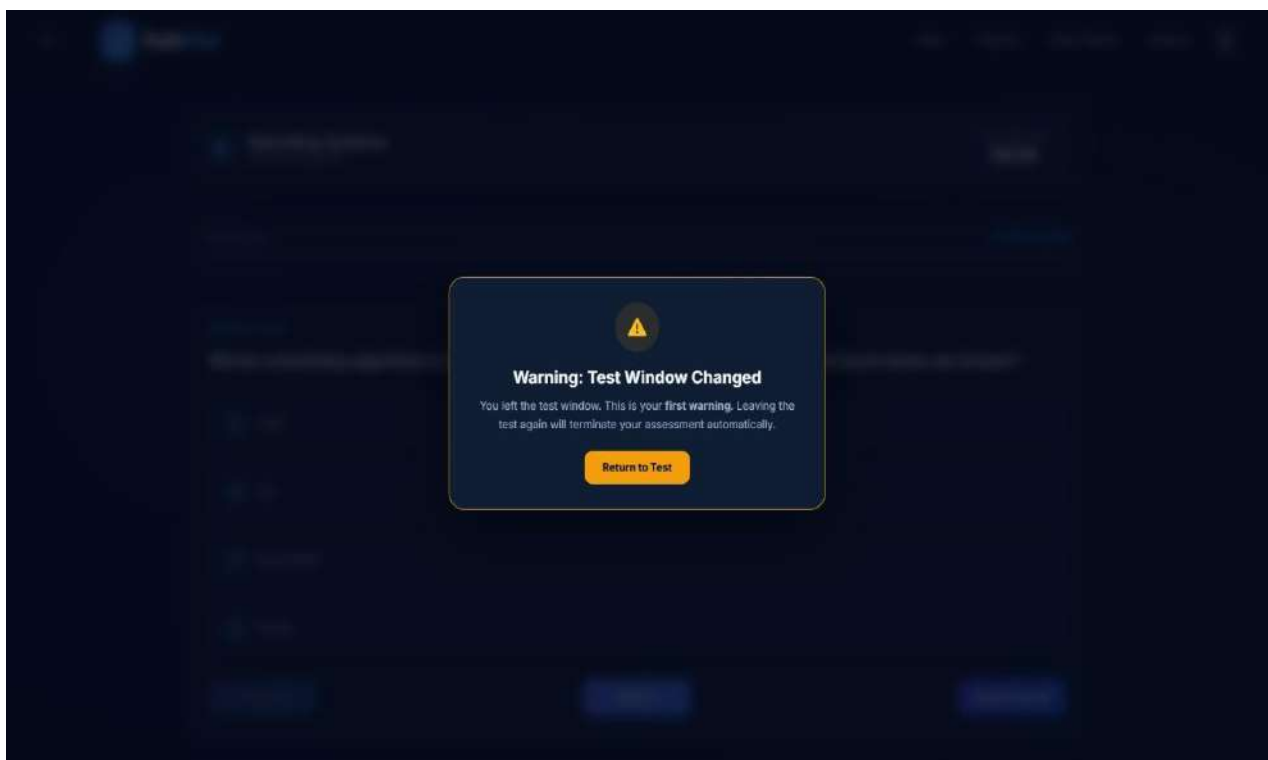


FIGURE 10. Coding Assessment – Test Window Warning

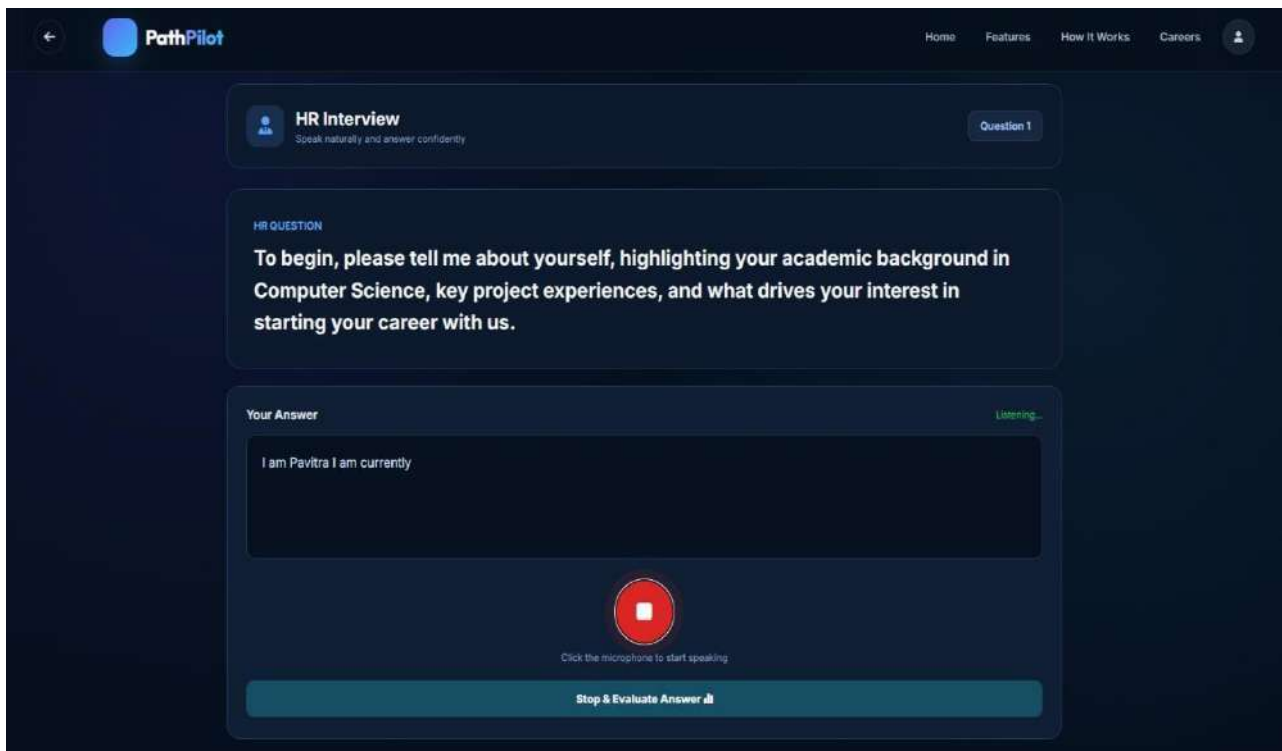


FIGURE 11. HR Interview – Interactive Interview Session

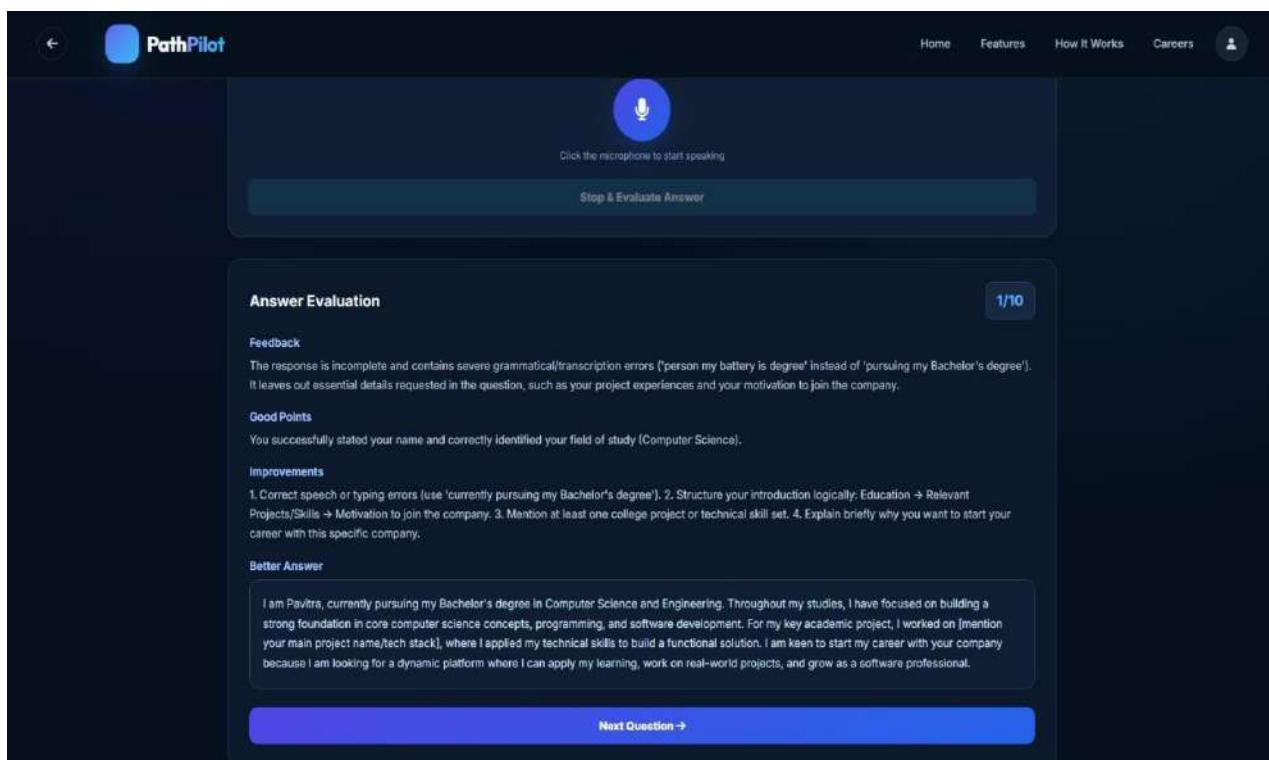


FIGURE 12. AI-Powered Answer Evaluation

7. CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

PathPilot was developed as an AI-powered career guidance platform designed to support students in making informed career decisions.

The system integrates student profile management, an AI Career Guidance Chatbot, a Resume Analyzer, and a career knowledge base into a single web-based platform.

The AI Career Guidance Chatbot allows students to interact with the system and obtain guidance related to career paths, job roles, skills, certifications, companies, and learning opportunities.

The Resume Analyzer allows students to upload their resumes and receive personalized analysis based on both the resume and the student's stored profile.

A major feature of PathPilot is the connection between the student profile and the AI-based modules. This enables the system to provide personalized recommendations instead of providing identical generic suggestions to every student.

The Resume Analyzer can also distinguish between information missing from the resume and potential actual skill gaps by comparing resume content with the student's profile.

The integration of the career knowledge base provides additional information related to companies, packages, certifications, hiring information, and career mapping.

Therefore, PathPilot demonstrates how artificial intelligence, student profile information, structured career knowledge, and resume analysis can be combined to create a personalized career guidance platform for students.

7.2 FUTURE WORK

Although PathPilot provides important career guidance and resume analysis features, several enhancements can be implemented in future versions.

The possible future improvements include:

- Real-time job recommendation integration
- Advanced job description and resume matching
- Improved ATS analysis
- Resume version comparison
- Personalized learning roadmap generation
- Skill progress tracking
- Advanced interview preparation
- Course and certification recommendations
- Company-specific preparation guidance
- Real-time job market information
- More advanced career analytics

- Integration with external job portals

Future versions can also improve the personalization of the AI system by using more detailed student history and continuously updated career datasets.

These improvements can help PathPilot evolve from a career guidance application into a more comprehensive AI-powered student career development platform.

PUBLICATION STATUS

The PathPilot project has been prepared for dissemination through academic conferences.

The project has been applied/submitted to the following conferences:

1. Researcher Pub → Paper published; presentation scheduled for next Saturday.
2. ICCISS 2027 – KIET → Awaiting response.
3. ICADCML 2027 – NIT Puducherry → Awaiting response.
4. IISTJ Journal → Awaiting response.
5. IJPS Journal → Awaiting response.

The publication status may vary by conference depending on the respective review and decision process. Where available, submission confirmation is retained as evidence of application. This report therefore records the conferences to which the project has been applied rather than presenting the submissions as accepted publications.

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PathPilot: An AI-Powered Career Recommendation and Resume Intelligence System for Engineering Students

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Abstract— Choosing the career path and making a good resume is something that all Undergraduate Computer Science and Engineering students have to deal with. The current resume checkers give advice and do not take into account a student's background, skills or goals. Career counselling services offer fixed and not advice. This paper presents PathPilot. A web-based tool that helps with career guidance and resume analysis. It uses chat-based career counselling and resume review to solve these problems and offer a personal service. The tool was built using a Flask backend, an SQLite database, a Jinja2/JavaScript frontend and the Gemini large language model. Unlike resume checkers PathPilot creates a context based on the student's profile and skills including skills and knowledge projects, certifications and career goals. This context is then combined with the resume text. Given to the large language model. The model gives back a JSON object that has a score ATS score, resume quality score, target role score, analysis of skills gaps, keyword coverage before and after suggestions to keep the facts and get the right format and recommendations for roles and companies based on the career knowledge graph. The tool clearly shows the difference between skills the student didn't mention and skills the student doesn't have. It also avoids making up any data, achievements or certifications. This paper explains why PathPilot was created its design, database structure resume processing, design and how it was built. It talks about the benefits compared to resume checkers and gives a general evaluation of the system. More studies, with users are needed for data.

Keywords— career guidance system, resume analysis, applicant tracking system, skill-gap analysis, generative AI, Flask, natural language processing, student profiling

I. INTRODUCTION

Every year many students who study Computer Science and Engineering go into the job market not knowing what to do. These students face challenges when trying to find the right areas where they can use their skills and reach their career goals. They also worry about whether their resumes will pass the tests companies use to select candidates. Career counseling at schools is too general. Doesn't help each student individually. At the time the programs that check resumes only look at the information people give and rate them based on certain rules. These rules might not match what the company is looking for in terms of education, goals and long-term plans.

Two problems are still not solved by the tools that're available. First the feedback on resumes is not personal. A tool that checks resumes cannot know that a person already has a skill unless the skill is written down as a keyword. It also can't tell the difference between a skill that is missing from the resume and a skill that the person never learned. Second career help and resume reviews are usually tools. This means students have to look at the advice from a chatbot about careers and the resume feedback separately and then try to make sense of both on their own.

PathPilot was created to solve both of these problems in one place. The system keeps a student profile that includes educational background, skills with levels of experience, projects, certifications and career interests. This profile acts as the base for any resume check or career question. The platform brings together four features that're usually separate: (i) managing a career profile that is specific to each user, including education skills with levels, projects and certifications; (ii) a chatbot that gives career help; (iii) a

resume checker that is made for the systems companies use and gives several scores; and (iv) checking for missing skills and giving advice to improve the resume without losing any important information.

The rest of the paper is arranged like this. Section II looks at research. Section III explains the system and how it was designed. Section IV describes how the system works overall. Section V talks about the database setup. Section VI explains how the resume-checking process works. Section VII covers how the AI models are used and how the questions are written. Section VIII gives details about how the system was built. Section IX shows the results and what was learned. Section X ends with a conclusion and ideas, for what could come.

II. RELATED WORK

Automated resume screening has been widely studied in the context of Applicant Tracking Systems (ATS). These systems usually depend on keyword matching, expression-based section detection or simple parsing to determine if a resume should move forward to human review [1]. While they are good at handling numbers of applications keyword-based ATS systems often penalize resumes that use different terms to describe real experience. They also offer no explanation for why a resume was rejected.

The rise of transformer-based language models has greatly improved how machines understand text. Models like BERT [2] showed that contextual language representations could outperform methods like bag-of-words or TF-IDF on many natural language tasks. The attention mechanism introduced in [3] powers modern large language models today including those used for open-ended reasoning and text generation. Recently multimodal generative models such as Google's Gemini family [4] and instruction-tuned models similar to GPT-3 [5] have made it possible to get structured outputs like JSON-formatted evaluations from a natural-language prompt. This is done without training a classifier for every specific task.

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PathPilot follows a layered, monolithic web-application architecture, chosen for its simplicity of deployment and ease

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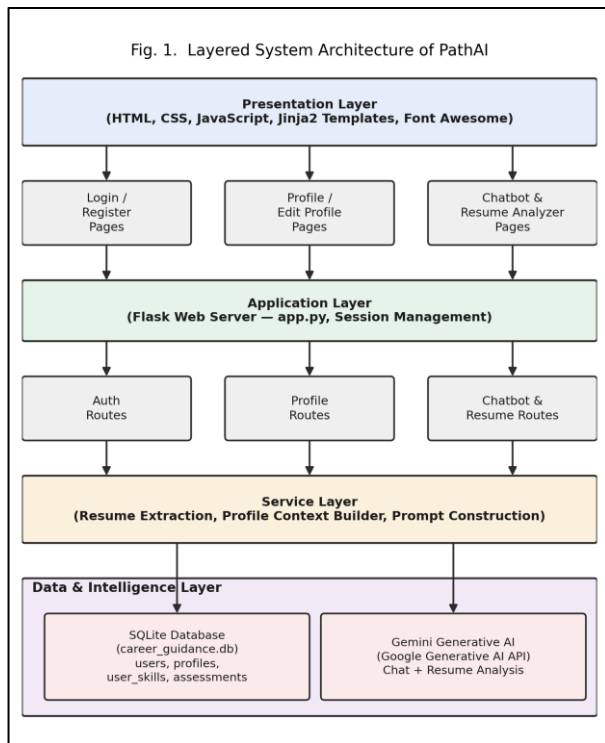


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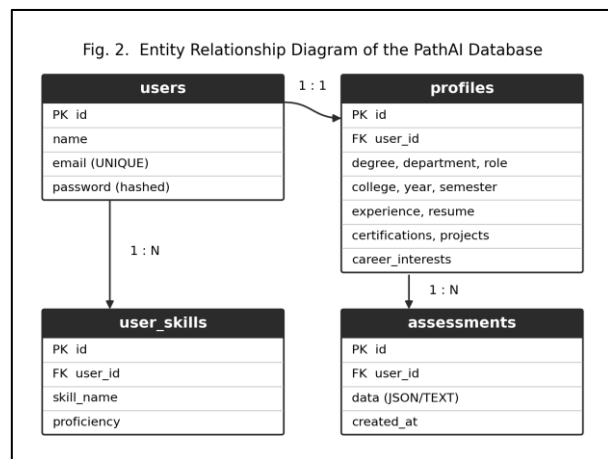


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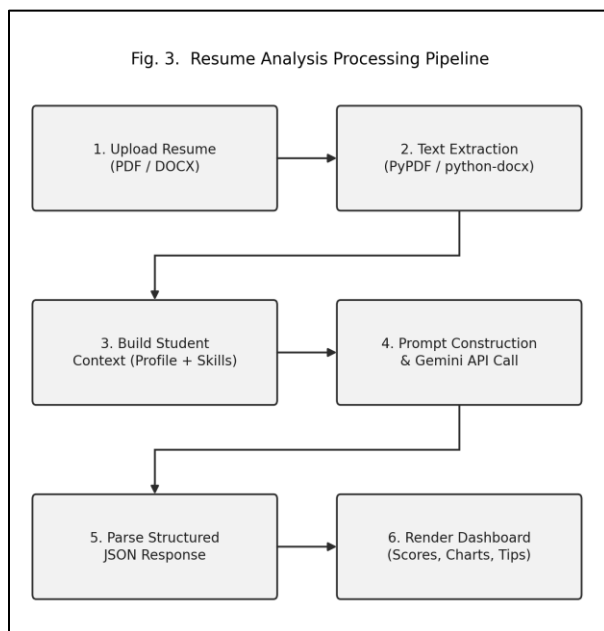
The assessments table holds a JSON snapshot of every analysis carried out for a user along with its creation

timestamp. This lets the user keep a history of their resume reviews for reference.

All foreign-key relationships link to user(id). All service-layer queries are filtered by the session identifier of the logged-in user so that data for students stays isolated

VI. RESUME ANALYSIS PIPELINE

The Resume Analyzer accepts resumes in PDF and DOCX formats through a POST request to the /resume/analyze endpoint. Users must be logged in to use this endpoint; otherwise they will get an HTTP 401 response, for access. The full resume processing pipeline is shown in Fig. 3.



Once the user uploads and checks their resume the system reads the text from the document. For a PDF the system reads each page one after the other. For a DOCX file it uses python-docx. If the system cannot read any text it will show an error. All the text that the system reads is joined with the user's verified profile. The profile lists qualifications, skills, projects, certifications, career interests, an optional target role and the job description. The system sends this data to the Gemini model. Waits for a JSON reply. The reply holds scores, strengths, problems, skill assessment, keywords, skill gaps, rewrite suggestions and recommendations. Finally the backend checks the results. Shows them on a clear dashboard.

VII. AI INTEGRATION AND PROMPT DESIGN

A. Grounding the Model in Verified Student Data

One design choice I want to highlight is to stop the model from guessing about a resume. Before asking any question or giving feedback the chatbot and resume analyzer give the

model a picture of the logged-in student's profile. This way the model looks first at the student's verified data in the database not at the resume text.

B. Explicit Factual-Integrity Constraints

The system prompt gives the model a set of instructions. Important the model must not create fake achievements, numbers, certificates or work history. It must not ask the student to guess their skills. Also if a skill is missing from the resume the model must tell whether the skill is just not mentioned or the student truly lacks it because each case needs a response. Likewise the model should not suggest tools that are only trendy. Those tools must match the student's goals and current skills. Instead the model should recommend skills that fit the job the student wants match the student's level fill skill gaps and avoid overusing keywords. These rules are not generic. They are custom for each student based on their needs and the current analysis. Also when giving resume tips the model may rewrite text a bit. The new version must keep all claims. Use different words. When recommending companies the model should look up the career knowledge base to avoid naming companies. The model's scores must be useful. Not promise a job. Important if the model finds conflicting data it must not fix it itself. It should point it out to the user for fixing.

C. Structured Output Contract

than letting the model format the answer in HTML or Markdown, the chatbot and resume analyzer ask Gemini to give structured data. For the chatbot this means a human-readable message that fits the chat interface. For the resume analyzer it means following the JSON schema. Thus the front end can show the data cleanly using badges, progress bars, charts and other tools of plain text from the model. This approach is very helpful for PathPilot giving an experience, than a simple chat.

VIII. IMPLEMENTATION

A. Authentication and Session Management

User registration and login are handled with securely hashed passwords. Once a user is authenticated, their identifier and display name are saved in the Flask session (session['user_id'], session['user_name']). Before accessing any secure routes, each request is checked against an active session to ensure that the correct user-specific information is retrieved. A new user must complete their profile before they can access the homepage whereas an existing user with a completed profile will be routed straight to the dashboard.

This overall process for the authentication and routing is summarized in Fig. 4.

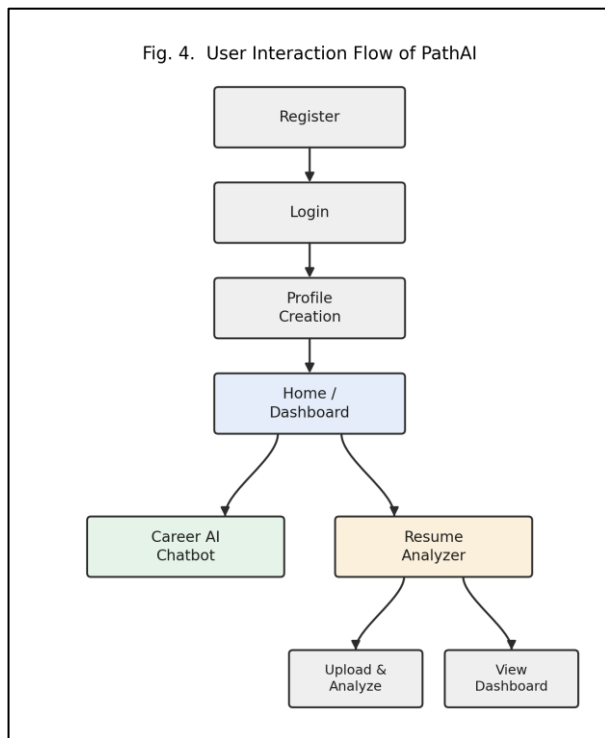


Fig. 4. User interaction flow of PathPilot.

B. Profiles

In the profiles section there are two templates that are used. These are called profile.html. In the profile.html the student can see their information, education, skills, experience, projects, certifications, career interests and extracurricular activities. On the hand the edit_profile.html is only available when a student wants to make some changes to their profile. This is a way to make sure that the profile is not changed directly on the view page. When a student logs in the navigation bar has an icon with a dropdown menu. This dropdown menu shows the "View Profile" option of a logout option. This is done to stop students from leaving the page.

C. Conversational Career Chatbot

This conversational career chatbot can be used by the students to ask their questions in a way. The questions are then sent as a JSON payload through the POST /chat endpoint. A JSON response is also received. The chatbot is designed to answer questions about careers, jobs, required skills, relevant companies, ways to learn and tech- areas of interest. These areas include software development, data and automation testing. The chat interface has chat history, a typing indicator and suggestion chips. These chips help first-time users ask the questions.

D. Resume Analyzer Dashboard

The analyzer dashboard is the area where a student's resume is checked. Students first upload their resume in PDF format

for it to be analyzed. This dashboard has a resume preview, suggestions for improvement and a downloadable report. Students can reach the Resume Analyzer through a card on the landing page. This card opens up in a dashboard-style interface, not a simple upload form. They can upload their resume choose a target role if they want add a job description and start the analysis. When the returned JSON is shown the frontend displays the score, ATS score resume quality score and target-role match. These are shown using some form of indicators. The strengths and important issues are listed. The keyword analysis is shown with matched and missing keyword badges. The skill comparison is shown in a bar-style diagram. This diagram compares the resume-evidenced score to the expected one.

E. Career and Company Recommendation

Career and company suggestions are included in the JSON response, as the resume analysis. This means there is no recommendation system. The prompt that tells Gemini to create suggestions includes the student profile that was used to create the resume. It also includes the resume content if it was provided and the target role if it was specified. The JSON response that is generated has the recommended_skills, recommended_roles and recommended_companies fields. As explained in Section VII-B the recommended companies are taken from a career knowledge database that is carefully prepared. This means that they are not made up by the model but are taken from data. In the way the recommended roles are focused on the student's skill gaps and their interests. They are not just based on what's popular. This method makes sure that both the resume analysis and the career suggestions are based on the student profile data. This data is included in the prompt of the Gemini API request.

IX. RESULTS AND DISCUSSION

The modules, which we have been implementing (authentication, personal profile storage, skills storage with proficiency levels, career guidance chatbot, resume analytics backend with PDF/DOCX parsing), have been tested for their ability to store and process data specific to each user without interfering with other users' data, which is critical for this platform to maintain multi-user isolation. Table I below is an example of the skill comparison output for an example student profile parsed by the resume analyzer module. It demonstrates the level of proficiency indicated by the resume text for each skill, compared to the level required for a specific position. The most common source of discrepancy, prompting skill gap and learning recommendation reports in the chatbot is not the presence or absence of specific keywords in the resume, but rather the level of proficiency indicated in the resume text, if it differs from the required level, as demonstrated for the Statistics skill below.

TABLE I

Sample Skill Comparison Output

Skill	Resume Score	Expected Score
Python	80	90
SQL	70	85
Git	60	80
Statistics	30	80

Compared to traditional keyword-based ATS checkers, the profile-grounded approach used in PathPilot helps distinguish between two situations that may appear the same to a normal resume scanner: a skill the student has not learned and a skill the student possesses but has not mentioned in the resume. This distinction is important because the required action is different in each case: a missing skill requires a learning recommendation, while an existing skill not mentioned only requires a suggestion for improved resume wording. Combining profile-based analysis with clearly defined factual-integrity rules for the language model, PathPilot therefore offers resume improvement suggestions and skill recommendation based on the student's actually declared information instead of content generated or invented by the model.

The structured JSON response format also provides an additional advantage in that it allows for a single AI response to power multiple dashboard elements, such as the score indicators, keyword badges, skill-comparison charts, and prioritized improvement lists, removing the need for the frontend to interpret or reformat free-form AI-generated text and directly supporting the project's requirement that the student should not be presented with raw and unstructured AI output.

X. CONCLUSION AND FUTURE WORK

This paper introduces PathPilot. PathPilot is an AI-based platform that offers career consulting and resume feedback. PathPilot brings together student profiling, a career chatbot and resume scoring modules in a single Flask-based web application.

The main feature of PathPilot is grounded resume analysis. Unlike tools PathPilot does not try to score the resume on its own. PathPilot instead uses the students profile declared skills, proficiency and career interests as the reference. The PathPilot model is told not to invent achievements or company names. The PathPilot model is penalized if it ignores keywords. This approach keeps the focus on improvements. The PathPilot model outputs are in JSON format. This format lets PathPilot build a dashboard. The dashboard shows scores, skill gaps, keyword statistics and resume suggestions.

The main limitations of the study involve evaluation and the knowledge base. In Section IX the assessment of the PathPilot chatbot and resume analysis is mainly qualitative

and focuses on architecture. There was no controlled experiment to measure how accurate the scoring, skill extraction and recommendations are. A future research option is to run a user study. The study would compare the results of grounded resume analysis with labels from a comparable dataset. It could also evaluate how the score changes when using a skills profiler (see Section VII). Also although the PathPilot knowledge base covers professions it has not been updated with recent developments in the job market. This lack of updates threatens the relevance of the recommendations.

Future research directions suggested by the authors include updating the PathPilot knowledge base. They also suggest adding versions of a students resume. This would allow a resume-history view. Enable tracking of scoring improvements with each analysis by using the assessments table. Another idea is to use retrieval-augmented generation. This would help recommend companies and roles from an external knowledge base. The approach would also reduce hallucination risks. Finally a user study with engineering students could be conducted. The study would evaluate how effective PathPilot is compared to resume checkers that were common, at the time.

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PathPilot: An AI-Powered Career Recommendation and Resume Intelligence System for Engineering Students

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Abstract— Choosing the career path and making a good resume is something that all Undergraduate Computer Science and Engineering students have to deal with. The current resume checkers give advice and do not take into account a student's background, skills or goals. Career counselling services offer fixed and not advice. This paper presents PathPilot. A web-based tool that helps with career guidance and resume analysis. It uses chat-based career counselling and resume review to solve these problems and offer a personal service. The tool was built using a Flask backend, an SQLite database, a Jinja2/JavaScript frontend and the Gemini large language model. Unlike resume checkers PathPilot creates a context based on the student's profile and skills including skills and knowledge projects, certifications and career goals. This context is then combined with the resume text. Given to the large language model. The model gives back a JSON object that has a score ATS score, resume quality score, target role score, analysis of skills gaps, keyword coverage before and after suggestions to keep the facts and get the right format and recommendations for roles and companies based on the career knowledge graph. The tool clearly shows the difference between skills the student didn't mention and skills the student doesn't have. It also avoids making up any data, achievements or certifications. This paper explains why PathPilot was created its design, database structure resume processing, design and how it was built. It talks about the benefits compared to resume checkers and gives a general evaluation of the system. More studies, with users are needed for data.

Keywords—career guidance system, resume analysis, applicant tracking system, skill-gap analysis, generative AI, Flask, natural language processing, student profiling

I. INTRODUCTION

Every year many students who study Computer Science and Engineering go into the job market not knowing what to do. These students face challenges when trying to find the right areas where they can use their skills and reach their career goals. They also worry about whether their resumes will pass the tests companies use to select candidates. Career counseling at schools is too general. Doesn't help each student individually. At the time the programs that check resumes only look at the information people give and rate them based on certain rules. These rules might not match what the company is looking for in terms of education, goals and long-term plans.

Two problems are still not solved by the tools that're available. First the feedback on resumes is not personal. A tool that checks resumes cannot know that a person already has a skill unless the skill is written down as a keyword. It also can't tell the difference between a skill that is missing from the resume and a skill that the person never learned. Second career help and resume reviews are usually tools. This means students have to look at the advice from a chatbot about careers and the resume feedback separately and then try to make sense of both on their own.

PathPilot was created to solve both of these problems in one place. The system keeps a student profile that includes educational background, skills with levels of experience, projects, certifications and career interests. This profile acts as the base for any resume check or career question. The platform brings together four features that're usually separate: (i) managing a career profile that is specific to each user, including education skills with levels, projects and certifications; (ii) a chatbot that gives career help; (iii) a

resume checker that is made for the systems companies use and gives several scores; and (iv) checking for missing skills and giving advice to improve the resume without losing any important information.

The rest of the paper is arranged like this. Section II looks at research. Section III explains the system and how it was designed. Section IV describes how the system works overall. Section V talks about the database setup. Section VI explains how the resume-checking process works. Section VII covers how the AI models are used and how the questions are written. Section VIII gives details about how the system was built. Section IX shows the results and what was learned. Section X ends with a conclusion and ideas, for what could come.

II. RELATED WORK

Automated resume screening has been widely studied in the context of Applicant Tracking Systems (ATS). These systems usually depend on keyword matching, expression-based section detection or simple parsing to determine if a resume should move forward to human review [1]. While they are good at handling numbers of applications keyword-based ATS systems often penalize resumes that use different terms to describe real experience. They also offer no explanation for why a resume was rejected.

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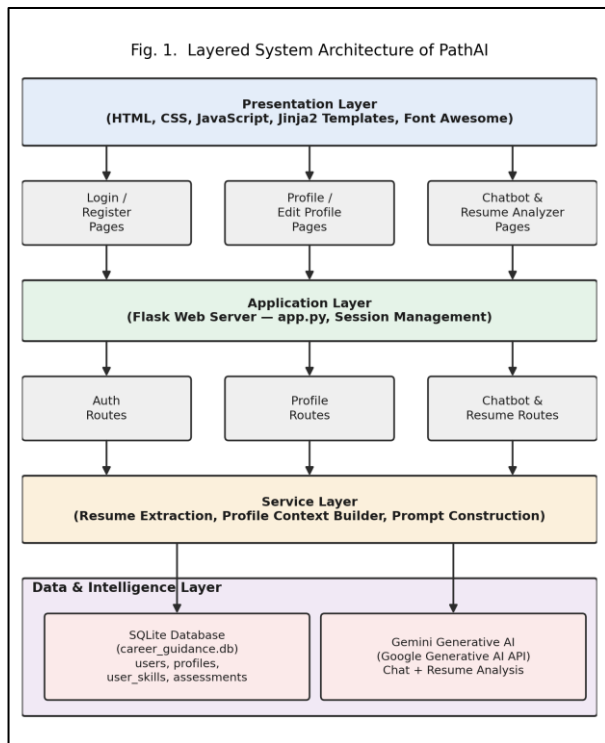


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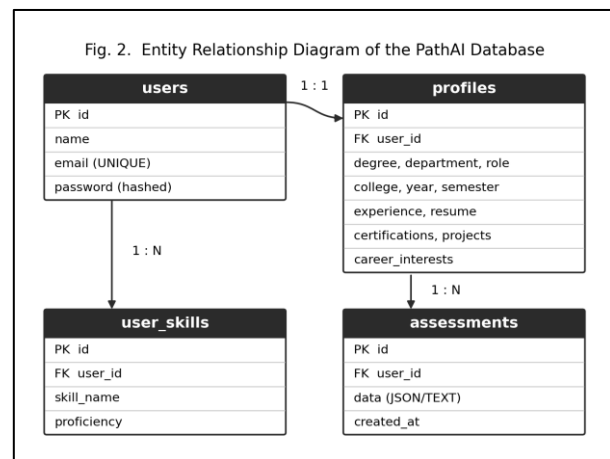


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Skills are kept in an one-to-many table instead of being stored in a single text field inside the profile. Each entry contains the skill name and the proficiency level, which can be Beginner, Intermediate or Advanced. This lets the application consider the actual level of expertise for a given skill than just checking if a keyword appears in a candidate's resume. As a result the accuracy of skill discovery during resume parsing improves.

The assessments table stores a JSON representation of each assessment performed on the user's data with the creation timestamp. Each record includes the skill_name and the proficiency level, such as Beginner, Intermediate or Advanced. This separation lets PathPilot differentiate between the candidate's level of expertise and mere keywords, which is essential for distinguishing declared skills from keywords during resume parsing.

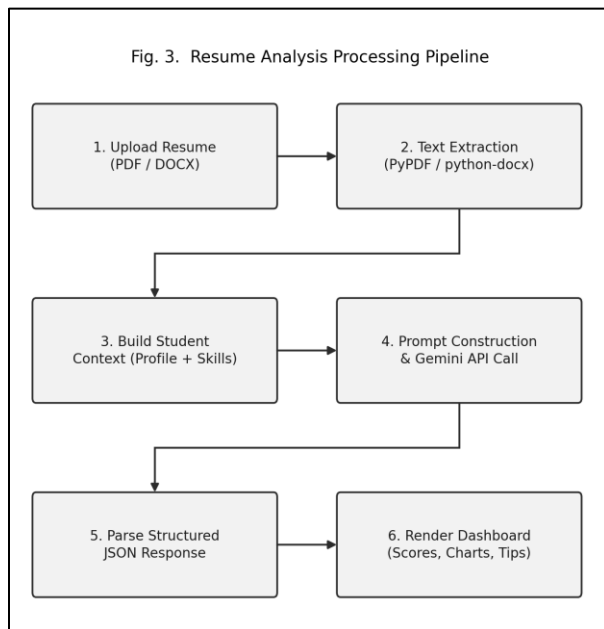
The assessments table holds a JSON snapshot of every analysis carried out for a user along with its creation

timestamp. This lets the user keep a history of their resume reviews for reference.

All foreign-key relationships link to user(id). All service-layer queries are filtered by the session identifier of the logged-in user so that data for students stays isolated

VI. RESUME ANALYSIS PIPELINE

The Resume Analyzer accepts resumes in PDF and DOCX formats through a POST request to the /resume/analyze endpoint. Users must be logged in to use this endpoint; otherwise they will get an HTTP 401 response, for access. The full resume processing pipeline is shown in Fig. 3.



Once the user uploads and checks their resume the system reads the text from the document. For a PDF the system reads each page one after the other. For a DOCX file it uses python-docx. If the system cannot read any text it will show an error. All the text that the system reads is joined with the user's verified profile. The profile lists qualifications, skills, projects, certifications, career interests, an optional target role and the job description. The system sends this data to the Gemini model. Waits for a JSON reply. The reply holds scores, strengths, problems, skill assessment, keywords, skill gaps, rewrite suggestions and recommendations. Finally the backend checks the results. Shows them on a clear dashboard.

VII. AI INTEGRATION AND PROMPT DESIGN

A. Grounding the Model in Verified Student Data

One design choice I want to highlight is to stop the model from guessing about a resume. Before asking any question or giving feedback the chatbot and resume analyzer give the

model a picture of the logged-in student's profile. This way the model looks first at the student's verified data in the database not at the resume text.

B. Explicit Factual-Integrity Constraints

The system prompt gives the model a set of instructions. Important the model must not create fake achievements, numbers, certificates or work history. It must not ask the student to guess their skills. Also if a skill is missing from the resume the model must tell whether the skill is just not mentioned or the student truly lacks it because each case needs a response. Likewise the model should not suggest tools that are only trendy. Those tools must match the student's goals and current skills. Instead the model should recommend skills that fit the job the student wants match the student's level fill skill gaps and avoid overusing keywords. These rules are not generic. They are custom for each student based on their needs and the current analysis. Also when giving resume tips the model may rewrite text a bit. The new version must keep all claims. Use different words. When recommending companies the model should look up the career knowledge base to avoid naming companies. The model's scores must be useful. Not promise a job. Important if the model finds conflicting data it must not fix it itself. It should point it out to the user for fixing.

C. Structured Output Contract

than letting the model format the answer in HTML or Markdown, the chatbot and resume analyzer ask Gemini to give structured data. For the chatbot this means a human-readable message that fits the chat interface. For the resume analyzer it means following the JSON schema. Thus the front end can show the data cleanly using badges, progress bars, charts and other tools of plain text from the model. This approach is very helpful for PathPilot giving an experience, than a simple chat.

VIII. IMPLEMENTATION

A. Authentication and Session Management

User registration and login are handled with securely hashed passwords. Once a user is authenticated, their identifier and display name are saved in the Flask session (session['user_id'], session['user_name']). Before accessing any secure routes, each request is checked against an active session to ensure that the correct user-specific information is retrieved. A new user must complete their profile before they can access the homepage whereas an existing user with a completed profile will be routed straight to the dashboard.

This overall process for the authentication and routing is summarized in Fig. 4.

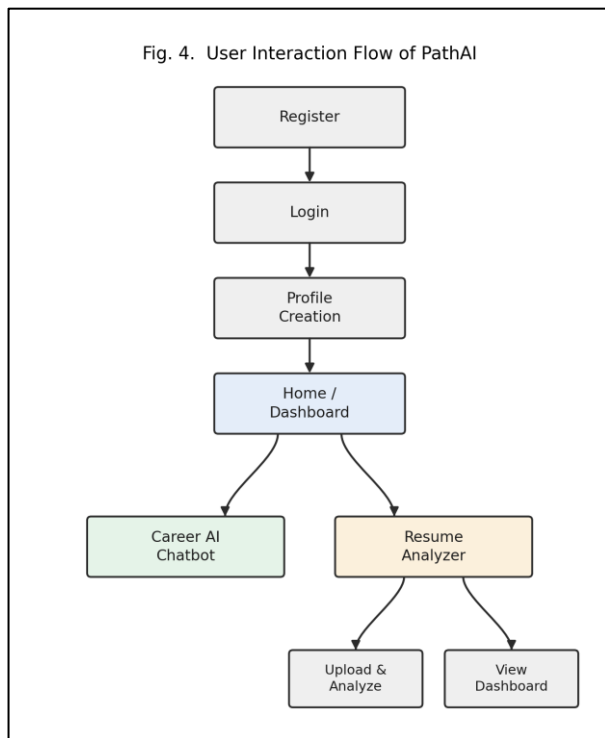


Fig. 4. User interaction flow of PathPilot.

B. Profiles

In the profiles section there are two templates that are used. These are called profile.html. In the profile.html the student can see their information, education, skills, experience, projects, certifications, career interests and extracurricular activities. On the hand the edit_profile.html is only available when a student wants to make some changes to their profile. This is a way to make sure that the profile is not changed directly on the view page. When a student logs in the navigation bar has an icon with a dropdown menu. This dropdown menu shows the "View Profile" option of a logout option. This is done to stop students from leaving the page.

C. Conversational Career Chatbot

This conversational career chatbot can be used by the students to ask their questions in a way. The questions are then sent as a JSON payload through the POST /chat endpoint. A JSON response is also received. The chatbot is designed to answer questions about careers, jobs, required skills, relevant companies, ways to learn and tech- areas of interest. These areas include software development, data and automation testing. The chat interface has chat history, a typing indicator and suggestion chips. These chips help first-time users ask the questions.

D. Resume Analyzer Dashboard

The analyzer dashboard is the area where a student's resume is checked. Students first upload their resume in PDF format

for it to be analyzed. This dashboard has a resume preview, suggestions for improvement and a downloadable report. Students can reach the Resume Analyzer through a card on the landing page. This card opens up in a dashboard-style interface, not a simple upload form. They can upload their resume choose a target role if they want add a job description and start the analysis. When the returned JSON is shown the frontend displays the score, ATS score resume quality score and target-role match. These are shown using some form of indicators. The strengths and important issues are listed. The keyword analysis is shown with matched and missing keyword badges. The skill comparison is shown in a bar-style diagram. This diagram compares the resume-evidenced score to the expected one.

E. Career and Company Recommendation

Career and company suggestions are included in the JSON response, as the resume analysis. This means there is no recommendation system. The prompt that tells Gemini to create suggestions includes the student profile that was used to create the resume. It also includes the resume content if it was provided and the target role if it was specified. The JSON response that is generated has the recommended_skills, recommended_roles and recommended_companies fields. As explained in Section VII-B the recommended companies are taken from a career knowledge database that is carefully prepared. This means that they are not made up by the model but are taken from data. In the way the recommended roles are focused on the student's skill gaps and their interests. They are not just based on what's popular. This method makes sure that both the resume analysis and the career suggestions are based on the student profile data. This data is included in the prompt of the Gemini API request.

IX. RESULTS AND DISCUSSION

The modules, which we have been implementing (authentication, personal profile storage, skills storage with proficiency levels, career guidance chatbot, resume analytics backend with PDF/DOCX parsing), have been tested for their ability to store and process data specific to each user without interfering with other users' data, which is critical for this platform to maintain multi-user isolation. Table I below is an example of the skill comparison output for an example student profile parsed by the resume analyzer module. It demonstrates the level of proficiency indicated by the resume text for each skill, compared to the level required for a specific position. The most common source of discrepancy, prompting skill gap and learning recommendation reports in the chatbot is not the presence or absence of specific keywords in the resume, but rather the level of proficiency indicated in the resume text, if it differs from the required level, as demonstrated for the Statistics skill below.

TABLE I

Sample Skill Comparison Output

Skill	Resume Score	Expected Score
Python	80	90
SQL	70	85
Git	60	80
Statistics	30	80

Compared to traditional keyword-based ATS checkers, the profile-grounded approach used in PathPilot helps distinguish between two situations that may appear the same to a normal resume scanner: a skill the student has not learned and a skill the student possesses but has not mentioned in the resume. This distinction is important because the required action is different in each case: a missing skill requires a learning recommendation, while an existing skill not mentioned only requires a suggestion for improved resume wording. Combining profile-based analysis with clearly defined factual-integrity rules for the language model, PathPilot therefore offers resume improvement suggestions and skill recommendation based on the student's actually declared information instead of content generated or invented by the model.

The structured JSON response format also provides an additional advantage in that it allows for a single AI response to power multiple dashboard elements, such as the score indicators, keyword badges, skill-comparison charts, and prioritized improvement lists, removing the need for the frontend to interpret or reformat free-form AI-generated text and directly supporting the project's requirement that the student should not be presented with raw and unstructured AI output.

X. CONCLUSION AND FUTURE WORK

This paper introduces PathPilot. PathPilot is an AI-based platform that offers career consulting and resume feedback. PathPilot brings together student profiling, a career chatbot and resume scoring modules in a single Flask-based web application.

The main feature of PathPilot is grounded resume analysis. Unlike tools PathPilot does not try to score the resume on its own. PathPilot instead uses the students profile declared skills, proficiency and career interests as the reference. The PathPilot model is told not to invent achievements or company names. The PathPilot model is penalized if it ignores keywords. This approach keeps the focus on improvements. The PathPilot model outputs are in JSON format. This format lets PathPilot build a dashboard. The dashboard shows scores, skill gaps, keyword statistics and resume suggestions.

The main limitations of the study involve evaluation and the knowledge base. In Section IX the assessment of the PathPilot chatbot and resume analysis is mainly qualitative

and focuses on architecture. There was no controlled experiment to measure how accurate the scoring, skill extraction and recommendations are. A future research option is to run a user study. The study would compare the results of grounded resume analysis with labels from a comparable dataset. It could also evaluate how the score changes when using a skills profiler (see Section VII). Also although the PathPilot knowledge base covers professions it has not been updated with recent developments in the job market. This lack of updates threatens the relevance of the recommendations.

Future research directions suggested by the authors include updating the PathPilot knowledge base. They also suggest adding versions of a students resume. This would allow a resume-history view. Enable tracking of scoring improvements with each analysis by using the assessments table. Another idea is to use retrieval-augmented generation. This would help recommend companies and roles from an external knowledge base. The approach would also reduce hallucination risks. Finally a user study with engineering students could be conducted. The study would evaluate how effective PathPilot is compared to resume checkers that were common, at the time.

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